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# On the role of partial least squares in path analysis for the social sciences

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ABSTRACT

We describe the current and potential future roles for partial least squares (PLS) algorithms in path analyses, guided by recent advances in envelope theory. After reviewing the present debate and establishing a context, we conclude that, depending on specific objectives, PLS methods have considerable promise, but that their full potential, while reachable, is not now being realized. The future developments necessary for achieving their full potential in the social sciences are clear and doable, albeit demanding. A critique of covariance-based structural equation modeling (CB-SEM), as it relates to PLS, is given as well. Technical details are available in the appendix.

## 1. Introduction

Path modeling is a standard way of representing social science theories. The validation of a path model often involves inference about concepts like "customer satisfaction" or "competitiveness" for which there are no objective measurement scales. Since such concepts cannot be measured directly, multiple surrogates, which may be called indicators, observed or manifest variables, are used to gain information about them indirectly. One role of a path diagram is to provide a visual representation of the relationships between the concepts represented by latent variables and the indicators. A fully executed path diagram is in effect a model that can be used to guide subsequent analysis, rather like an algebraic model in statistics. A path model is commonly translated to a structural equation model (SEM) for analysis.

There are two common paradigms for studying a structural equation model: partial least squares (PLS-SEM) with origins tracing back to Wold (1982), and covariance-based analysis (CB-SEM) which is based on the work of Jöreskog (1978). These have for decades been used for estimation and inference, but advocates of these two paradigms now seem at loggerheads over their relative advantages and disadvantages, due in part to publications (eg. Rönkkö & Evermann, 2013; Rönkkö, McIntosh, & Edwards, 2016b) that are highly critical of PLS-SEM methods. The methods that we reference as PLS-SEM are also referred to as PLS path modeling and the PLS approach to structural equation modeling.

### *1.1. The PLS-SEM vs CB-SEM debate*

Much discussion has taken place in the social sciences literature about the role of PLS-SEM in path modeling (Akter, Wamba, & Dewan, 2017; Evermann & Rönkkö, 2021; Goodhue, Lewis, & Thompson, 2023; Henseler et al., 2014; Rönkkö & Evermann, 2013; Rönkkö et al., 2016b; Russo & Stol, 2023; Sarstedt, Hair, Ringle, Thiele, & Gudergan, 2016; Sharma, Liengaard, Sarstedt, Hair, & Ringle, 2023). Rönkkö et al. (2016b) asserted that "…the majority of claims made in PLS literature should be categorized as statistical and methodological myths and urban legends", and recommended that the use of "…PLS should be discontinued until the methodological problems explained in [their] article have been fully addressed". In response, proponents of PLS-SEM appealed for a more equitable assessment. Henseler and his nine coauthors (Henseler et al., 2014) tried to reestablish a constructive discussion on the role PLS-SEM in path analysis while arguing that, rather than moving the debate forward, Rönkkö and Evermann (2013) created new myths. McIntosh, Edwards, and Antonakis (2014) attempted to resolve the issues separating Rönkkö and Evermann (2013) and Henseler et al. (2014) in an 'even-handed manner,' concluding by lending their support to Rigdon's (2012) recommendation that steps be taken to divorce PLS-SEM path analysis from its competitors by developing methodology from a purely PLS perspective. See also Petter (2018), Petter and Hadavi (2021) for subsequent calls for balance. The appeal for balance was found wanting by Rönkkö et al. (2016b), who restated and reinforced the views of Rönkkö and Evermann (2013), and by the Editors of *The Journal of Operations Management*, who in 2015 established a policy of desk-rejecting papers that use PLS-SEM (Guide & Ketokivi, 2015). To our knowledge, the position of *J. Operations Management* is an outlier as no other journal has taken such an extreme posture, and many journals have published special issues devoted to PLS-SEM.

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Recently Evermann and Rönkkö (2021) tempered their ardent criticism of PLS-SEM when writing "…to ensure that [Information Systems (IS)] researchers have up-to-date methodological knowledge of PLS if they decide to use it". Although dressed up a bit to highlight new developments, their criticism are essentially the same as those they leveled in previous writings (eg. Rönkkö & Evermann, 2013; Rönkkö et al., 2016b). There was much give-and-take in the subsequent discussion, both for and against PLS-SEM. For instance, Sharma et al. (2023) expressed matter-of-factly that the PLS-SEM claims leveled by Evermann and Rönkkö (2021) are misleading, extraordinary and questionable. They then set about "…to bring a positive perspective to this debate and highlight the recent developments in PLS that make it an increasingly valuable technique in IS and management research in general". While the various factions seem to have shifted their positions somewhat, this most recent debate did not aid our understanding materially.

There is a substantial literature that bears on this debate and we have not attempted a comprehensive review (Rönkkö et al., 2016b cites about 150 references). Nevertheless, our impression is that the preponderance of articles rely mostly on intuition and simulations to support sweeping statements about intrinsically mathematical/statistical issues without adequate supporting theory. And disagreements have arisen even when care is taken. For instance, McIntosh et al. (2014) took issue with Henseler et al.'s (2014) description of a "composite factor model" arguing that it is instead a "common factor model" and that a composite factor model is of the formative type. It seems to us that adequately addressing the methodological issue in path analysis requires in part avoiding ambiguity by employing a degree of context-specific theoretical specificity.

In this article we use the acronym PLS to designate the partial least squares methods that stem from the theoretical foundations established by Cook, Helland, and Su (2013). These link with early work by Wold (Galadi, 1988) and cover chemometrics applications (Cook & Forzani, 2020, 2021; Martens & Næs, 1989) as well as a host of subsequent methods, particularly the recent PLS methods by Cook, Forzani, and Liu (2023) for simultaneous reduction of predictors and responses in multivariate regression that is relevant to the analysis of the path models in Fig. 2.1. We rely on these foundations to elucidate PLS-SEM. Cook et al. (2013) showed that PLS is an envelope method (see also Cook, 2018) and consequently it inherits the structure of envelopes, which allows for a sharper understanding of the potential for PLS in path analyses.

### 1.2. Goal

We primarily intend to elucidate PLS-SEM methods and to describe the future potential of PLS methods in path analyses. We discuss only CB-SEM as a backdrop for PLS-SEM. We restrict discussion to a reflective path model so the context is clear, as it seems to us that many claims are made without adequate foundations. The reflective path model that we use provides a foundation that allows us to state clearly results that may carry over to more intricate settings, as discussed in Section 7. And we are able to avoid terms and phrases that do not seem to be understood in the same way across the community of path modelers (eg. McIntosh et al., 2014; Sarstedt et al., 2016). A method that performs well in our relatively straightforward context may show similar performance in more complicated settings. But we would not expect a method that performs poorly in our context to do better in more complex path models.

Although we have written extensively about PLS algorithms (Cook & Forzani, 2017, 2019, 2020, 2021; Cook et al., 2023, 2013), relatively little detailed information seems available on their use in PLS-SEM. We relied on technical details from the book by Lohmöller (1989) and articles by Wold (1975, 1982) and by Dijkstra (1983), Dijkstra and Henseler (2015a, 2015b). Dijkstra and Lohmöller were both students of Wold. Output from our implementation of their description of a PLS-SEM estimator agreed with output from an algorithm by Rönkkö, McIntosh, Antonakis, and Edwards (2016a), which supports our assessment of the method. We confine our discussion largely to issues encountered in Wold's first stage algorithm (Wold, 1982, Section 1.4.1). If an implementation of Wold's first stage does not produce sound results, subsequent stages may not be serviceable.

### 1.3. Outline

We set the context for our study in Section 2, including the reflective path diagrams that will inform our discussion and the measurement of association between the latent constructs in Section 2.3 that was influenced by Rigdon (2012). Estimation is considered in Section 3. We focus on PLS-SEM and CB-SEM until Section 4 where we turn to envelopes and PLS. Simulation results are given in Section 6. The technical details in the appendix were developed specifically for the path diagrams described in Section 2. Further research is needed to extend these results to broader issues that may be encountered in more complicated models. A possible starting point for such extensions is sketched in Appendix G. Our broad conclusions are given in Section 7.

## 2. Reflective path models

### 2.1. Reflective path diagrams

The upper and lower path diagrams presented in Fig. 2.1 give two typical reflective path models (eg. Lohmöller, 1989) that will be used to inform our discussion. In each diagram $Y = (Y_j, j = 1, \dots, r)$ and $X = (X_j, j = 1, \dots, p)$ denote observable random vectors, whose elements $Y_j$ and $X_j$ we call indicators, that are assumed to provide reflective information about underlying real latent constructs $\xi, \eta$. This is indicated by the arrows in Fig. 2.1 leading from the constructs to the indicators. Boxed variables are observable while circled variables are not. There is no restriction from a psychometric perspective that indicators reflect only one construct. Our restriction to real constructs is intended to simplify and focus the discussion on crucial topics. It does not notably limit our conclusions, which can be extended, but with notable overhead, to multivariate constructs.

An important goal in this setting is to infer the association between $\eta$ and $\xi$, as indicated by the double-headed curved paths between them. We take this to be a general objective, and it is one focal point of this article. Each indicator is affected by an unobservable error $\epsilon$. The absence of paths between these errors indicates that, in the upper path diagram, they are independent or at least uncorrelated conditional on the latent variables. They are allowed to be conditionally dependent in the lower path diagram, as indicated by the double-arrowed paths that join them. We refer to the lower model as the correlated error (CORE) model and to the upper model as the uncorrelated error (UNCORE) model. The models in Fig. 2.1 cover simulation results by Rönkkö et al. (2016b, Figure 1), which is one reason for adopting them.

The path diagrams presented in Fig. 2.1 represent two fundamentally different reflective path models. In the UNCORE model, the latent construct $\xi$ accounts for *all* of the association between the $X$ indicators. These indicators are independent (or perhaps just uncorrelated) among sampled units with the same value of $\xi$. For instance, we might model $X_j = A_j + B_j\xi + \epsilon_j$, where $A_j, B_j$ are regression coefficients and the errors $\epsilon_j$, $j = 1, \dots, p$, are uncorrelated. Aside from scaling variables, this is the same as the model used by Dijkstra and Henseler (2015a, eq's (1)–(3)) as a basis for their consistent and asymptotically normal PLS-SEM estimators for linear structural equations, results that are partly responsible for the recent developments highlighted by Evermann and Rönkkö (2021, Section 4). This setting places a burden on social scientists who have the daunting task of selecting indicators that reflect only the latent construct and random error — and nothing else. The CORE model in Fig. 2.1 is the same as the UNCORE model, save the important exception

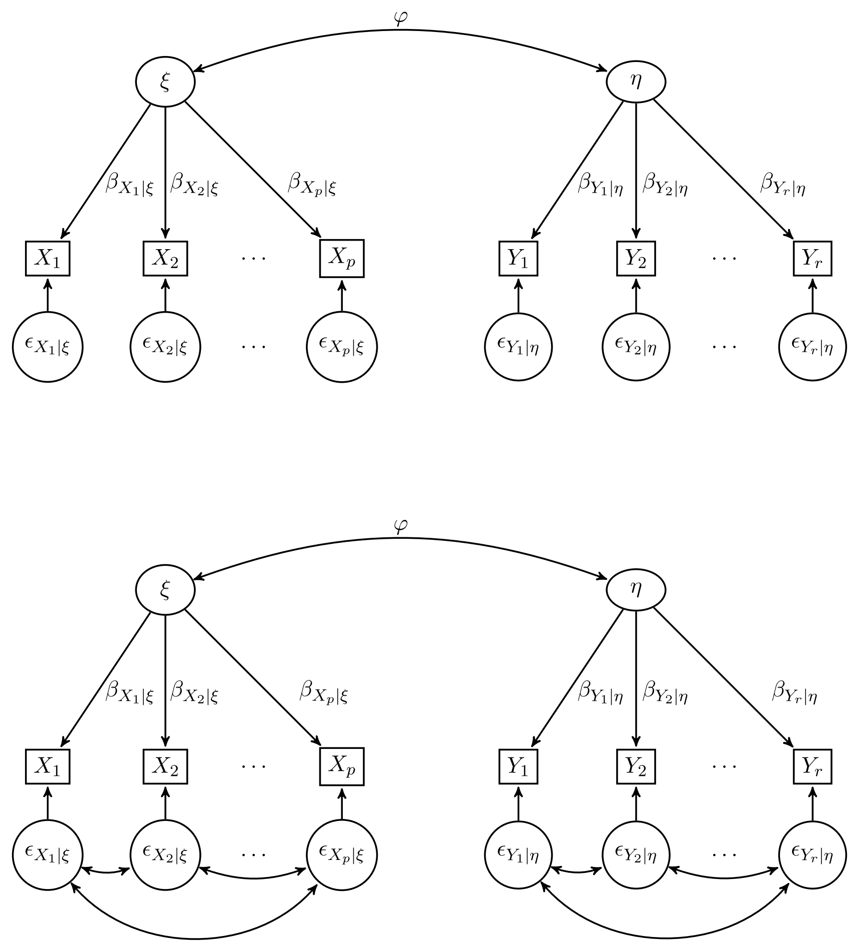


**Fig. 2.1.** Reflective path diagrams relating two latent constructs $\xi$ and $\eta$ with their respective univariate indicators, $X_1, \ldots, X_p$ and $Y_1, \ldots, Y_r$. $\varphi$ denotes either $\mathrm{cor}(\xi, \eta)$ or $\mathrm{cor}\{E(\xi \mid X), E(\eta \mid Y)\}$. We refer to the upper model as the uncorrelated error model (UNCORE) and to the lower model as the correlated error model (CORE).

that the errors are allowed to be correlated. The CORE model is much less restrictive than the UNCORE model.

The models of Fig. 2.1 occur frequently in the literature on path models. For instance, the UNCORE model is essentially the same as that described by Wold (1982, Fig. 1), who also commented that uncorrelated errors are not essential for the PLS estimation algorithm. They were used by Henseler et al. (2014, Fig. 1) in their discussion of Rönkkö and Evermann (2013), asserting that Rönkkö and Evermann (2013) conclusions are based on the UNCORE model. They reasoned further that the UNCORE model rarely holds in applied research, as argued much earlier by Schönemann and Wang (1972). Nevertheless, the UNCORE model is the basis for Dijkstra and Henseler's studies (Dijkstra, 1983; Dijkstra & Henseler, 2015a, 2015b) of the asymptotic properties of PLS-SEM. We have not seen a theoretical analysis of PLS-SEM under the CORE model.

### 2.2. Data, objectives and notation

We denote population means using the symbol $\mu$; for example, $\mu_X$ and $\mu_Y$ denote the population means of the indicator vectors. We denote the linear regression functions for the models in Fig. 2.1 using conditional mean notation: $\mathrm{E}(\eta \mid Y)$, $\mathrm{E}(\xi \mid X)$, $\mathrm{E}(Y \mid \eta)$ and $\mathrm{E}(X \mid \xi)$ are regression functions that are material to our discussion. The algebraic model implied by Fig. 2.1 is shown in Appendix A.

The only observables in the path models of Fig. 2.1 are $X$ and $Y$. We assume that the data are independent observations on $(X, Y)$. Accordingly, all available information stems from the joint distribution of $X$ and $Y$. Since we are assuming normality, this joint distribution is fully characterized by its mean vector and covariance matrix $\Sigma_{(X,Y)}$ with sample version $S_{(X,Y)}$:

$$\Sigma_{(X,Y)} = \begin{pmatrix} \Sigma_X & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_Y \end{pmatrix}, \; S_{(X,Y)} = \begin{pmatrix} S_X & S_{XY} \\ S_{YX} & S_Y \end{pmatrix}.$$

where, for random vectors $A$ and $B$, $\Sigma_A = \mathrm{var}(A)$ with sample version $S_A$ and $\Sigma_{AB} = \mathrm{cov}(A, B)$, $\Sigma_{BA} = \Sigma_{AB}^T$ with sample versions $S_{AB}$ and $S_{BA}$. We use the notation $\Sigma_{A|B} = \mathrm{var}(A \mid B)$ to denote the conditional variance matrix of $A$ given $B$.

Normality per se is not necessary. However, certain implications of normality like linear regressions with constant variances are needed. Assuming normality overall avoids the need to provide a list assumptions, which might obscure the overarching points about the role of partial least squares.

### 2.3. Measuring association between $\xi$ and $\eta$

Regardless of the model selected from Fig. 2.1, the association between $\eta$ and $\xi$ has been measured in the path modeling literature via the correlation between regression functions $\mathrm{cor}\{\mathrm{E}(\eta \mid Y), \mathrm{E}(\xi \mid X)\}$, or via the correlation $\mathrm{cor}(\eta, \xi)$ (See, for example, Lohmöller, 1989 and Dijkstra, 1983, Section 4.3). These measures are estimative focal points of this article.

The correlation measures $\Psi := \mathrm{cor}\{\mathrm{E}(\eta \mid Y), \mathrm{E}(\xi \mid X)\}$ and $\mathrm{cor}(\eta, \xi)$ can reflect quite different views of a reflective setting. The marginal correlation $\mathrm{cor}(\eta, \xi)$ implies to us that $\eta$ and $\xi$ represent concepts that are uniquely identified regardless of any subjective choices made regarding the reflective indicators. Two investigators studying the same nominal constructs would naturally be estimating the same correlation even if they selected different indicators. In contrast, $\Psi$, the correlation between population regressions, suggests a conditional view: $\eta$ and $\xi$ exist only by virtue of the indicators that are selected as reflective of their properties and so attributes of $\eta$ and $\xi$ cannot be claimed without citing the corresponding $(Y, X)$. Two investigators studying the same concepts could understandably be estimating different correlations if they selected different indicators. For instance, the latent concept "happiness" might reflect in part a combination of happiness at home and happiness at work. Two sets of indicators that weight these happiness subtypes differently might yield different correlations with a second latent construct, say "generosity". As pointed out by a referee, when gauging a construct by using a questionnaire, the questions together with the respondents' understanding of them define the construct's meaning and thus define the concept, which may not be the same as the target concept. Rigdon (2012) questioned whether constructs like $\xi$ and $\eta$ are necessarily the same as the theoretical concepts they are intended to capture or whether they are approximations of the theoretical concepts allowed by the selected indicators. If the latter view is taken, then an understanding of the constructs cannot be divorced from the specific indicators selected for the study.

Further, Rigdon, Becker, and Sarstedt (2019) argued that there is *always* a gap between a targeted concept like consumer satisfaction or trustworthiness and its representation as a construct via selected indicators. Attempts to rectify a concept through a specific set of indicators must naturally reflect the researchers' vision, insights and ideology, with the unavoidable implication that $\mathrm{cor}(\eta, \xi)$ is a researcher-specific parameter. Two independent researcher groups studying the same concepts, but using different indicators, will inherently be dealing with distinct correlations $\mathrm{cor}(\eta, \xi)$. Whether they arrive at the same substantive conclusions is a related issue that can also be relevant (Rigdon, Sarstedt, & Ringle, 2017), but is beyond the scope of this article. The correlation $\Psi$ brings these issues to the fore; its explicit dependence on the indicators manifests important transparency and candor, and it facilitates the reification of a concept, although it does not directly address discrepancies between the target concept and construct.

The diagrams in Fig. 2.1 can be well described as instances of operational measurement theory in which concepts are defined in terms of the operations used to measure them (e.g., Bridgman, 1927; Hand, 1996). In operationalism there is no assumption of an underlying objective reality; the concept being pursued is precisely that which is being measured, so concept and construct are the same. In the alternate measurement theory of representationalism, the measurements are distinct from the underlying reality. There is a universal reality underlying the concept of length regardless of how it is measured. But a universal understanding of 'happiness' must rely on unanimity over the measurement process.

## 3. Estimation

### 3.1. Setup

The constructs $\eta$ and $\xi$ are not well-defined since any non-degenerate linear transformation of them lead to an equivalent model and, in consequence, it is useful to introduce harmless constraints to facilitate estimation, inference and interpretation. To this end, we consider two types of constraints:

$$\text{Regression constraints}\quad \mu_\xi = \mu_\eta = 0 \text{ and } \mathrm{var}\{\mathrm{E}(\xi \mid X)\} = \mathrm{var}\{\mathrm{E}(\eta \mid Y)\} = 1,$$
$$\text{Marginal constraints}\quad \mu_\xi = \mu_\eta = 0 \text{ and } \sigma^2_\xi = \sigma^2_\eta = 1,$$

where $\sigma^2_{(\cdot)}$ denotes the variance of the indicated real variable. The regression constraints fix the variances of the regression functions $\mathrm{E}(\xi \mid X)$ and $\mathrm{E}(\eta \mid Y)$ at 1 and consequently $\Psi = \mathrm{cov}(\mathrm{E}(\xi \mid X), \mathrm{E}(\eta \mid Y))$. The marginal constraints fix the marginal variances of $\xi$ and $\eta$ at 1 and so $\mathrm{cor}(\xi, \eta) = \mathrm{cov}(\xi, \eta)$. It seems to us that the choice of constraints is mostly a matter of taste: We show in Proposition D.3 that the choice has no impact on the end result. The marginal constraints were used by Dijkstra and Henseler (2015a) in their development of consistent PLS-SEM estimators, but these particular constraints should not affect the end result, either.

### 3.2. Estimation of $\mathrm{cor}(\eta, \xi)$

The CB-SEM estimator of $\mathrm{cor}(\eta, \xi)$ is based on maximum likelihood estimation from the normal model with the marginal constraints. However, identifiability issues arise without additional restrictions on the model. In particular, as discussed in Proposition D.2 of Appendix D, at least one identified off-diagonal element of the conditional variance matrix $\Sigma_{Y|\eta}$ must be set to 0 and at least one identified off-diagonal element of the conditional variance matrix $\Sigma_{X|\xi}$ must be set to 0 (Dijkstra, 1983, Section 4.1). It is frequently assumed that these covariance matrices are diagonal (eg. Dijkstra & Henseler, 2015a); we have not encountered any applications where less restrictive conditions are imposed. In other words, the usual CB-SEM estimator of $\mathrm{cor}(\eta, \xi)$ is appropriate only under the UNCORE model of Fig. 2.1. We have already commented on the limited applicability of the UNCORE model, in our opinion. We follow convention and assume that $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are diagonal matrices when estimating $\mathrm{cor}(\eta, \xi)$, understanding that this is a stand-in for assumptions described in Proposition D.2. We do not require $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ to be diagonal when estimating $\Psi$.

The CB-SEM estimator of $\mathrm{cor}(\eta, \xi)$ requires that the sample size be larger than the number of parameters; in particular, $S_X$ and $S_Y$ must be nonsingular. Rosseel's R package (Rosseel, 2012) for structural equation modeling, which we used for the CB-SEM estimator, gives error messages when sample covariance matrices are not positive definite.

### 3.3. Estimation of $\Psi = \mathrm{cor}\{\mathrm{E}(\eta \mid Y), \mathrm{E}(\xi \mid X)\}$

We show in Appendix B that the correlation $\Psi$ is identifiable without requiring restrictive assumptions on the off-diagonal elements of $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$, as does $\mathrm{cor}(\eta, \xi)$. We also show that the CORE model implies that the regressions of $Y$ on $X$ and $X$ on $Y$ are multivariate reduced rank models where the coefficient matrices are known to have rank one because $\eta$ and $\xi$ are real numbers. The rank can be larger than one when $\eta$ and $\xi$ are vectors.

#### 3.3.1. Maximum likelihood estimator

The multivariate reduced rank models derived in Appendix B enable us to demonstrate in Appendix C.1 that the maximum likelihood estimator $\widehat{\Psi}_{\mathrm{MLE}}$ of $\Psi$ is simply the first singular value $\upsilon_1$ of $C_{YX} = S_Y^{-1/2} S_{YX} S_X^{-1/2}$. Equivalently, it is the first sample canonical correlation between $X$ and $Y$.

In the terminology of Wold (1982), the path diagrams in Fig. 2.1 are characterized as mode A. Wold (1982, Section 5.2), pointed out that the first canonical correlation also arises under mode B (formative models), so the connection between canonical correlations and the analysis of Wold's two-block, two-LV soft models is not entirely novel.

#### 3.3.2. Moment estimator

Define

$$\widehat{\Psi}_{\mathrm{MT}} = \mathrm{tr}^{1/2}(S_{XY} S_Y^{-1} S_{YX} S_X^{-1}), \tag{3.1}$$

where $\mathrm{tr}$ denotes the trace operator. This moment estimator, which is derived in Appendix C.2, will not be as efficient as the maximum likelihood estimator under the model of Appendix A, but it might possess certain robustness properties.

#### 3.3.3. PLS-SEM estimator with marginal indicator scaling

We follow Dijkstra (1983), Dijkstra and Henseler (2015a) to compute one version of the PLS-SEM estimator of $\Psi$, as shown in detail in Appendix C.3. Briefly, first center and scale each indicator variable so it has mean 0 and standard deviation 1. Let $\tilde{X}$ and $\tilde{Y}$ denote the vectors of scaled indicators and let $\ell_1$ and $r_1$ denote the left and right singular vectors of the sample covariance matrix $S_{\tilde{Y}\tilde{X}}$ of the scaled indicators. Construct the proxy variables $\bar{\eta} = r_1^T \tilde{X}$ and $\bar{\xi} = \ell_1^T \tilde{Y}$. Then the PLS-SEM estimator with marginal indicator scaling $\widehat{\Psi}_{\text{PLS-MIS}}$ is the sample correlation between these proxies. Further study is needed to understand the general statistical properties of this estimator. This is clearly different from the maximum likelihood estimator $\widehat{\Psi}_{\text{MLE}}$, and we cannot see any clear rational for preferring it over other estimators discussed herein. Since the maximum likelihood estimator of $\Psi$ is asymptotically efficient, which requires normality, $\widehat{\Psi}_{\text{PLS-MIS}}$ is guaranteed to be no better than $\widehat{\Psi}_{\text{MLE}}$ under normality. It might be argued that $\widehat{\Psi}_{\text{PLS-MIS}}$ is serviceable when $S_X$ or $S_Y$ is singular, while $\widehat{\Psi}_{\text{MLE}}$ is not. That may be so, but we have seen no theoretical justifications and we see better potential options based on PLS, as described in Section 4.

The predictor scaling as used in the construction of $\widehat{\Psi}_{\text{PLS-MIS}}$ might be problematic because the singular vectors (equivalently, eigenvectors of $S_{\tilde{Y}\tilde{X}} S_{\tilde{Y}\tilde{X}}^T$ and $S_{\tilde{Y}\tilde{X}}^T S_{\tilde{Y}\tilde{X}}$), depend on the particular scaling used. The critical scaling issue was addressed by Hotelling (1933) in his development of principal components:

> The method of principal components can therefore be applied only if . . . a metric – a definition of distance – [is] assumed in the [p]-dimensional space, and not simply a set of axes . . .

This is strong counsel, as it warns that principal components (and therefore singular vectors) should be used *only if* a metric is defined, and presumably justified. Any conclusion drawn using PLS-SEM necessarily depend on the data-dependent scaling used. We know of no theoretical justification for the metric (marginal indicator scaling) used in PLS-SEM. However, there is theoretical justification for joint standardization.

#### 3.3.4. PLS-SEM estimator with joint indicator standardization

Another estimator $\widehat{\Psi}_{\text{PLS-JIS}}$ can be constructed by following the steps in Section 3.3.3, except joint indicator standardization (JIS) is used instead of marginal indicator scaling: $\tilde{X} = S_X^{-1/2}(X - \bar{X})$ and $\tilde{Y} = S_Y^{-1/2}(Y - \bar{Y})$. The sample correlation between the proxies is now the maximum likelihood estimator $\widehat{\Psi}_{\text{MLE}}$, as described in Section 3.3.1. More specifically, the theoretical justification in Appendix C.3 establishes that, under normality, $\widehat{\Psi}_{\text{MLE}} = \widehat{\Psi}_{\text{PLS-JIS}}$. We can see no rationale for preferring PLS-SEM with marginal indicator scaling $\widehat{\Psi}_{\text{PLS-MIS}}$ over the maximum likelihood estimator $\widehat{\Psi}_{\text{MLE}}$ when $S_X$ and $S_Y$ are nonsingular.

#### 3.3.5. Commentary

Of the four estimators discussed in Sections 3.3.1–3.3.4, two are maximum likelihood estimators and two are variations but all derive from the basic multivariate reduced rank model.

Under normality, the CB-SEM estimator and the PLS-SEM estimator $\widehat{\Psi}_{\text{PLS-JIS}}$ are both maximum likelihood estimators of their respective targets – $\text{cov}(\xi, \eta)$ and $\Psi$. The structural covariance restrictions needed for CB-SEM are not needed for PLS-SEM.

PLS is a dimension reduction method, as discussed in Section 5. In regression it provides for reducing the predictor vector without loss of information on the response (See, for example Cook, 2018). In path analyses its corresponding goal is to reduce the dimension of the indicator vector without loss of information on the construct. How it affects $R^2$, or its sensitivity to non-normality and measurement errors are ancillary issues The idea of estimating the correlations $\Psi$ and $\text{cor}(\xi, \eta)$ by using proxies seems to be a common theme in the path analysis literature, but we find the underlying motivation to be a bit elusive, as indicated previously in Section 3.3.3. Indeed, if joint indicator standardization is used then the proxy-based estimation of $\Psi$ results in the MLE, as described in Section 3.3.4. More generally, the regression functions $\text{E}(\xi \mid X)$ and $\text{E}(\eta \mid Y)$ can be seen as population proxies for $\xi$ and $\eta$. We elaborate on this in Appendix C.4.

Marginal predictor scaling is often used in standard linear regression models to facilitate interpretation. Scaling may be appropriate and useful here because the predictors are ancillary and properly treated as fixed constants. However, in PLS and in PLS-SEM the predictors and indicators are stochastic, their marginal distribution may carry relevant information about the constructs, and scaling may obscure some of that information. For this reason the scaling issue is crucial in the application of PLS-SEM. This issue arises also in the application of PLS in linear regression models (Cook, 2018; Cook et al., 2013; Cook & Su, 2016). Scaling is discussed further in Section 5.

### 3.4. Bias

Bias is a malleable concept, depending on the context, the estimator and the quantity being estimated.

If the goal is to estimate $|\text{cor}(\xi, \eta)|$ via maximum likelihood while assuming that $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are diagonal matrices, bias may not be a worrisome issue. Although maximum likelihood estimators are generally biased, the bias typically vanishes at a fast rate as the sample size increases. However, bias may be an issue when the sample size is not large relative to the number of parameters to be estimated while $S_X$ and $S_Y$ are still nonsingular. This issue is outside the scope of this article. A referee pointed out that the recent book edited by van de Schoot and Miočević (2020) could be relevant here.

If the goal is to estimate $|\Psi|$ without assuming diagonal covariance matrices and $n > \min(p, r) + 1$ then its maximum likelihood estimator, the first canonical correlation between $X$ and $Y$, can be used and again bias may not be a worrisome issue.

In some settings we may wish to use an estimator of $|\Psi|$ also as an estimator of $|\text{cor}(\xi, \eta)|$ without assuming diagonal covariance matrices. We show in Appendix E that under the marginal constraints, $\sigma_\xi^2 = \sigma_\eta^2 = 1$, we have

$$\Psi = [\text{var}\{E(\xi \mid X)\}\text{var}\{E(\eta \mid Y)\}]^{1/2}\text{cor}(\xi, \eta). \tag{3.2}$$

It follows that the relationship between $\Psi$ and $\text{cov}(\xi, \eta)$ depends on the variances of the regression functions, which are bounded: $\text{var}\{E(\xi \mid X)\} \le 1$ and $\text{var}\{E(\eta \mid Y)\} \le 1$, as shown in Appendix E. In consequence,

$$|\Psi| \le |\text{cor}(\eta, \xi)|. \tag{3.3}$$

Relationship (3.3) enables us to define a population bias in this case as the difference

$$0 \le |\text{cor}(\xi, \eta)| - |\Psi| \le 1,$$

which agrees with Dijkstra's (Dijkstra, 1983, Section 4.3) conclusion that PLS-SEM will underestimate $|\text{cor}(\xi, \eta)|$. See also Lohmöller (1989), Rönkkö et al. (2016b). As described in Appendix E, this bias will be small when $E(\text{var}(\xi \mid X))$ and $E(\text{var}(\eta \mid Y))$ are small, so $\xi$ and $\eta$ are well predicted by $X$ and $Y$. This may happen with a few highly informative indicators. It may also happen as the number of informative indicators increases, a scenario that is referred to as an *abundant* regression in statistics (Cook, Forzani, & Rothman, 2012; Cook et al., 2013), particular cases of which may correspond to consistency-at-large in PLS (See Appendix E). On the other extreme, if $\xi$ and $\eta$ are not well predicted by $X$ and $Y$ then it is possible to have $|\Psi|$ close to 0 while $|\text{cor}(\xi, \eta)|$ is close to 1, in which case the bias is close to 1. Like the assumption that $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are diagonal matrices, it may be effectively impossible to gain from the data support for claiming that $E(\text{var}(\xi \mid X))$ and $E(\text{var}(\eta \mid Y))$ are small.

In short, an estimator of $|\Psi|$ is also an estimator of a lower bound on $|\text{cor}(\xi, \eta)|$. The bias $|\text{cor}(\xi, \eta)| - |\Psi|$ will be small when the indicators are good predictors of the constructs.

## 4. How do envelopes fit?

Envelopes, which were introduced by Cook, Li, and Chiaromonte (2007) and developed for the multivariate linear model by Cook, Li, and Chiaromonte (2010), encompass a class of dimension reduction methods for increasing efficiency in multivariate analyses without altering traditional objectives. They serve to reshape classical methods by exploiting relationships that affect the accuracy of the results but are not recognized by classical methods. Envelope methods, which can result in efficiency gains equivalent to increasing the sample size many times over, operate generally by dividing the data into a part that is material to the goals of the analysis and a complementary part this is immaterial to those goals. The goals are then pursued effectively by using only the material part of the data. An introduction is available from Cook (2018), and envelope methods are now available for a wide variety of contexts, including Bayesian envelopes (Khare, Pal, & Su, 2016), reduced-rank regressions (Cook, Forzani, & Zhang, 2015), matrix-variate regressions (Ding & Cook, 2017), spatial modeling (Rekabdarkolaee, Wang, & Fluente, 2020), imaging (Park, Su, & Zhu, 2017), sparse regressions (Su, Zhu, Chen, & Yang, 2016), chemometrics regressions (Cook & Forzani, 2020), censored quantile regression (Zhao, Van Keilegom, & Ding, 2022), to mention a few of the envelope developments.

An envelope analysis of the path models displayed in Fig. 2.1 is based in part on dividing $X$ into $q \leq p$ material composites $X_R$ and $p-q$ complementary immaterial composites $X_{0R}$ needed to reconstruct $X$. This division is constructed so that

(a) $X_{0R}$ is independent of $\xi$, and thus it carries no marginal information about $\xi$.

(b) $X_{0R}$ is independent of $X_R$ given $\xi$, and so $X_{0R}$ cannot convey any information about $\xi$ due to an association with $X_R$.

These conditions guarantee that $X_{0R}$ does not reflect $\xi$ in any way; that is, $X_R$ can reflect $\xi$ but $X_{0R}$ cannot. In parallel, $Y$ is divided into $u < r$ composites $Y_R$ and $r-u$ complementary composites $Y_{0R}$ so that $Y_R$ reflects $\eta$ but $Y_{0R}$ does not. We think of $X_R$ as the material part of $X$ since it provides information on the construct. We think of $X_{0R}$ as the immaterial part of $X$ since variation in $X_{0R}$ provides no information on the construct either directly or indirectly via a correlation with $X_R$. The material and immaterial parts of $Y$ are defined similarly. There is also a close philosophical connection between this setup and the classical statistical notion of sufficiency in the sense that $X_R$ is sufficient for $\xi$. Since the reflective composites are unknown they must be estimated, including their dimensions $q$ and $u$.

The reflective composites can be estimated by using the multivariate regression model that derives from the path model in Fig. 2.1 (see Appendix F) and the corresponding likelihood-based envelope methods developed by Cook and Zhang (2015). These maximum likelihood envelope estimators of $X_R$ and $Y_R$ require optimizing a non-convex objective function (Cook, 2018; Cook et al., 2010), with $q$ and $u$ selected by using an information criterion like BIC, or by using cross validation or a holdout sample. The maximum likelihood estimator requires $S_X$ and $S_Y$ to be non-singular. It is unserviceable if either $S_X$ or $S_Y$ is singular. We refer to this method as envelope-based SEM, and we use the acronym EB-SEM. Also, we emphasize that EB-SEM is a maximum likelihood estimator that allows for dimension reduction, $q < p$ or $u < r$. The maximum likelihood estimator $\widehat{\Psi}_{\mathrm{MLE}}$ described in Section 3.3.1 does not allow for dimension reduction and it is the same as the EB-SEM estimator with $q = p$ and $u = r$.

The theoretical connections between envelopes and PLS established by Cook et al. (2013) implies immediately that PLS algorithms (Cook et al., 2023) are moment-based envelope methods that, like envelopes more generally, provide estimators of $X_R$ and $Y_R$. In PLS $q$ and $u$ are selected by using cross validation or a holdout sample. PLS algorithms do not require $S_X$ or $S_Y$ to be non-singular and so are adaptable to path models with relatively small sample sizes. We refer to this method as envelope-based PLS for SEMs, EB-PLS-SEM.

Likelihood-based envelopes and PLS methods including PLS-SEM are most effective and have the greatest potential in the CORE model. In the UNCORE model, $\mathrm{var}(X \mid \xi)$ and $\mathrm{var}(Y \mid \eta)$ must have identified off-diagonal elements that are known to be 0. Correspondingly, the material reflective variables $X_R$ and $Y_R$ can be used in MLE estimation for the UNCORE model provided $\mathrm{var}(X_R \mid \xi)$ and $\mathrm{var}(Y_R \mid \eta)$ have identified off-diagonal elements that are known to be 0. Requiring $\mathrm{var}(X \mid \xi)$, $\mathrm{var}(Y \mid \eta)$, $\mathrm{var}(X_R \mid \xi)$ and $\mathrm{var}(Y_R \mid \eta)$ to each have identified off-diagonal elements that are known to be 0 seems to be quite severe and, in consequence, we view likelihood-based envelopes, PLS and PLS-SEM as largely unserviceable for estimation in the UNCORE model.

Once $X_R$ and $Y_R$ have been estimated, whether by maximum likelihood or by PLS, they are treated as original data and estimation proceeds by using the methods described in Section 3. For instance, we pursue the correlation between the regression functions as $\Psi = \mathrm{cor}(E(\xi \mid X_R), E(\eta \mid Y_R))$ and its maximum likelihood estimator is then the first canonical correlation between the estimators of $X_R$ and $Y_R$. We do not recommend envelope methods for estimation of $\mathrm{cor}(\xi, \eta)$ as the assumption of diagonal covariance matrices can severely limit their utility. We see the assumption of diagonal covariance matrices as already imposing a drastic reduction in the number of parameters.

## 5. What is the role of PLS-SEM and what are its current limitations?

Aside from the practice of scaling the indicators by using their marginal standard deviations, we find that PLS-SEM is a special case, obtained by setting $q = u = 1$, of the simultaneous PLS method developed by Cook et al. (2023, Section 4.3). In other words, PLS-SEM assumes that all the information about $\xi$ that is available from $X$ is contained in one composite, so $X_R$ is a real variable. Similarly, $Y_R$ is also assumed to be real. Extending PLS-SEM to allow $q > 1$ and $u > 1$ then leads immediately to EB-PLS-SEM. In contrast to CB-SEM, EB-PLS-SEM does not require $S_X$ or $S_Y$ to be nonsingular. The statistical properties of the EB-PLS-SEM estimators when $n < p$ can be quite good (Cook & Forzani, 2017, 2019).

Appropriate scaling can be used to improve the performance of envelopes and PLS methods, generally, including PLS-SEM. Cook and Su (2013, 2016) developed likelihood-based envelope and PLS scaling methods, which turned out to be quite different from simply scaling by marginal standard deviations. In path analyses, the marginal standard deviations of the indicators are affected by both the material and immaterial variation in the indicators, and scaling by the marginal standard deviations can tend to obscure the material information. In consequence, scaling my marginal standard deviations is not advisable, generally. For details, see Section 1.1 of Cook and Su (2016) and Section A in its supplement.

In effect, PLS accounts for all *immaterial composites* found during estimation and this has the effect of reducing estimative and predictive variation. Methods that do not account for the immaterial composites are subject to extraneous variation that has the general effect of inflating estimative and predictive variation relative to PLS. The maximum likelihood estimators discussed in Sections 3.3.1 and 3.3.4 will be most appropriate when there are no immaterial composites, $X_{0R}$ and $Y_{0R}$ are nil, so all composites of the indicators are reflective of their constructs. However, these estimators may not be best when immaterial composites are present. In that case PLS and the maximum likelihood estimator based on $X_R$ and $Y_R$, come into play. In what follows, comments regarding EB-PLS-SEM apply also to PLS-SEM when it is assumed that $q = u = 1$ and no scaling is used.

The estimators that we have discussed are summarized in Table 1.

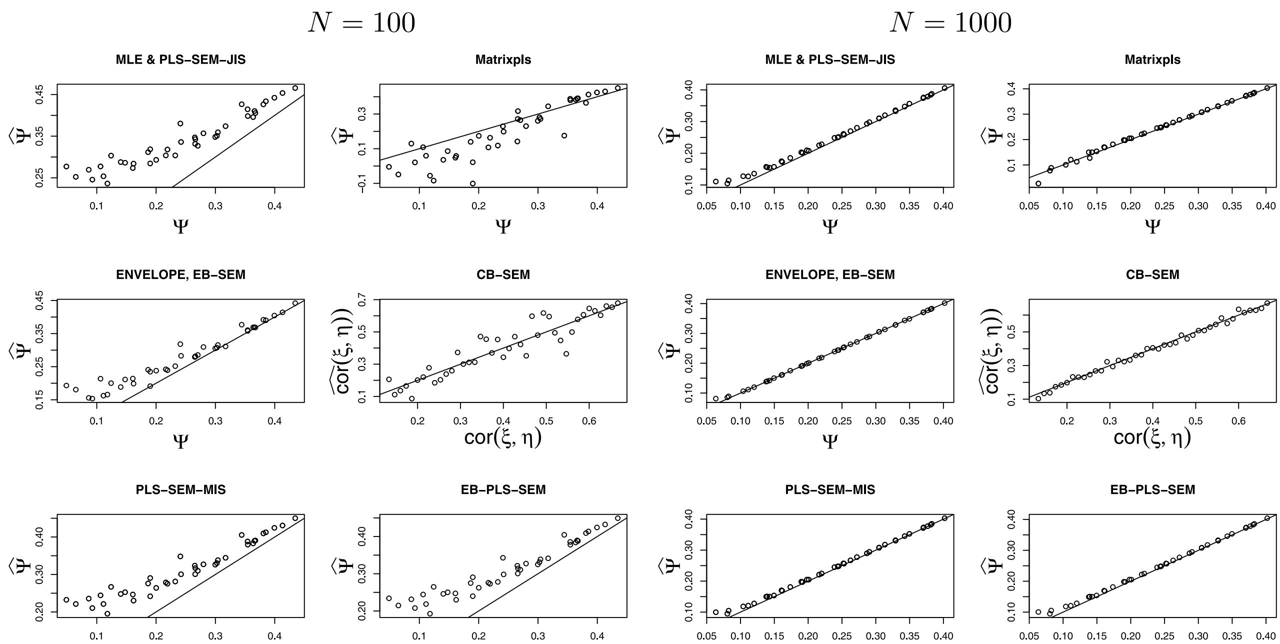


**Fig. 6.1.** Results of simulations with $\Sigma_{X|\xi} = \Sigma_{Y|\eta} = I_3$. Horizontal axes give the true correlations; vertical axes give the estimated correlations. Equality $x = y$ is represented by lines in the plots. MLE & PLS-SEM-JIS: as described in Section 3.3.1 and Section 3.3.4. Matrixpls: estimator from the code cited in Rönkkö et al. (2016b). ENVELOPE, EB-SEM: as described in Appendix F. CB-SEM: estimator from *lavaan* package. PLS-SEM-MIS: marginal standardization as described in Section 3.3.3. EB-PLS-SEM: This designates the envelope-based PLS estimator discussed in Section 4.

**Table 1**
Summary of serviceable estimation methods for the models of Fig. 2.1. None of the methods allow marginal indicator scaling, so PLS-SEM is not listed. $\widehat{\Psi}_{\mathrm{PLS-JIS}}$ is the same as $\widehat{\Psi}_{\mathrm{MLE}}$, and so is not listed explicitly. EB: envelope-based.

| $\mathrm{cor}(\xi, \eta)$ | $\Psi$ |
|---|---|
| CORE model, $S_X$ and $S_Y$ nonsingular | |
| CB-SEM<br>Unserviceable | $\widehat{\Psi}_{\mathrm{MLE}}, \widehat{\Psi}_{\mathrm{MT}}$, EB-SEM, EB-PLS-SEM |
| CORE model, $S_X$ or $S_Y$ singular | |
| CB-SEM<br>Unserviceable | EB-PLS-SEM |
| UNCORE model. $S_X$ and $S_Y$ must be nonsingular | |
| CB-SEM | $\widehat{\Psi}_{\mathrm{MLE}}, \widehat{\Psi}_{\mathrm{MT}}$ |

## 6. Simulation results

In this section, we include simulation results to support our general conclusions and to demonstrate what can happen. We do not attempt a comprehensive portrayal of the estimators. In all simulations, the constructs $\eta$ and $\xi$ are real random variables with $0 \le \mathrm{cor}(\eta, \xi) \le 2/3$. In Section 6.1 we use simulations in the manner of Rönkkö et al. (2016b) to illustrate that the PLS-SEM methods in the social sciences may do well in ideal settings where the SEM assumptions hold (Fig. 6.1). But in more realistic settings PLS-SEM methods may not be serviceable, while envelope methods, including EB-PLS-SEM, perform well (Figs. 6.2 and 6.3). The simulation results in Section 6.2 are intended to underscore the observation that CB-SEM and PLS-SEM methods may materially underestimate $\Psi$ and $\mathrm{cor}(\xi, \eta)$ when $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are not diagonal matrices, while the envelope-based methods continue to do well. We illustrate in Section 6.3 that CB-SEM and PLS-SEM methods may again exhibit substantial bias when there are multiple reflective composites.

### 6.1. Simulation results in the manner of Rönkkö et al. (2016b)

Rönkkö et al. (2016b) illustrated some of their concerns via simulations with $p = r = 3$ and sample size $n = 100$ under the marginal constraints with $0 \le \mathrm{cor}(\eta, \xi) \le 2/3$ and $\Sigma_{X|\xi} = \Sigma_{Y|\eta} = I_3$. We follow their general setup by simulating data with coefficient vectors for the linear regressions of $Y$ on $\eta$ and $X$ on $\xi$ set equal to $L = (8/10, -7/10, 6/10)^T$. Their choice for the covariance matrices is fortuitous because it implies that $u = q = 1$, so there is only one material composite of the indicators, which agrees with the construction of PLS-SEM. It follows from (3.2) that

$$\Psi = 0.5984 \times \mathrm{cor}(\eta, \xi), \tag{6.1}$$

which agrees with our general conclusion (3.3). Following Rönkkö et al. (2016b), we simulated data with sample sizes $N = 100$ and $N = 1000$ observations on $(X, Y)$ according to these settings with various values for $\mathrm{cor}(\eta, \xi)$. The estimators that we used are as follows:

MLE and PLS – SEM – JIS : This is the estimator described in Sections 3.3.1 and 3.3.4. We used our own code for computing.

Matrixpls : This R program, which was used to compute a version of PLS-SEM was provided to us by Mikko Rönkkö (Rönkkö et al., 2016b, Footnote 1).

EB – SEM : This envelope estimator was computed using the methods described in Cook and Zhang (2015). It is as discussed in Section 4.

CB – SEM : This estimator was discussed in Section 3.2. It was computed using the *lavaan* package based on Rosseel (2012).

PLS – SEM – MIS : This estimator is as discussed in Section 3.3.3. We computed it using our own code, but expected the estimates to be similar to those from Matrixpls. Discrepancies can arise because of differences in computing methods. Since Matrixpls is no longer supported in R, we turn to the R package cSEM to explain how differences might arise. The cSEM method ‘calculateWeightsPLS’ references Wold Wold (1975), who provided routines for computing weights. Imbedded in those routines are algorithms for computing eigenvectors. However, today there are better methods available for computing spectral decompositions. We did not use Wold’s routines, but instead computed

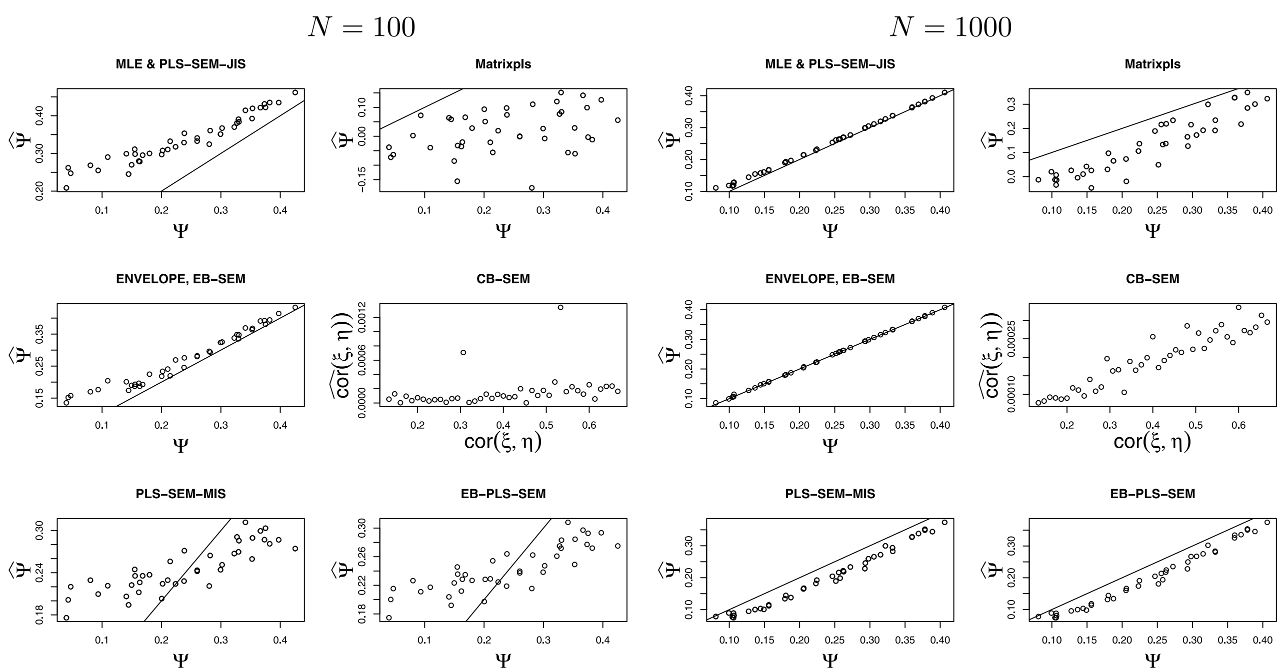


**Fig. 6.2.** Results of simulations with $\Sigma_{X|\xi} = \Sigma_{Y|\eta} = L(L^T L)^{-1} L^T + 3L_0 L_0^T$. Horizontal axes give the true correlations; vertical axes give the estimated correlations. The estimators are as described in the legend of Fig. 6.1.

spectral decompositions afresh based on the theoretical description provided by Wold (Wold, 1975, equations (61) and (62)) and later by Dijkstra (1983), Dijkstra and Henseler (2015a, 2015b).

EB – PLS – SEM This designates the PLS estimator discussed in Section 4. It was computed using the methods described in Cook et al. (2023, Table 2).

We can see from the results shown in Fig. 6.1, which are the averages over 10 replications (Rönkkö et al., 2016b used one replication), that at $N = 100$, the five estimators of $\Psi$ clearly overestimate for all $\mathrm{cor}(\eta, \xi) \in (0, 2/3)$, with EB-SEM doing a bit better than the others. Because the indicators are independent with constant variances conditional on the constructs, we did not expect much difference between PLS-SEM-JIS and PLS-SEM-MIS. None of these estimators made use of the fact that $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are diagonal matrices, as did CB-SEM. At $N = 1000$, the performance of EB-SEM is essentially flawless, while the other PLS-type estimators show a slight propensity to overestimate $\Psi$.

### 6.2. Non-diagonal $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$

Simulating with $\Sigma_{X|\xi} = \Sigma_{Y|\eta} = I_3$, as we did in the construction of Fig. 6.1, conforms to the standard assumption that these matrices be diagonal and partially accounts for the success of CB-SEM in Fig. 6.1. To illustrate the importance of these assumptions, we next took $\Sigma_{X|\xi} = \Sigma_{Y|\eta} = L(L^T L)^{-1} L^T + 8L_0 L_0^T$ with $L_0^T L_0 = I_2$ and $L^T L_0 = 0$. Otherwise, the settings are as described in Section 6.1, including the relationship between $\Psi$ and $\mathrm{cor}(\xi, \eta)$ given at (6.2). With this structure $X_R = L^T X$, $Y_R = L^T Y$, so the usual PLS-SEM assumption that $q = u = 1$ is still correct. $L_0^T Y$ and $L_0^T X$ satisfy the structural assumptions for PLS and envelopes. The results shown in Fig. 6.2 are quite different from those shown in Fig. 6.1 with two exceptions. At $N = 100$, EB-SEM is still the best with the MLE (equivalently PLS-SEM-JIS) coming in second. The performance of the other four estimators is terrible, as there seems to be little relationship between the estimator and the estimand. At $N = 1000$, the performance of EB-SEM is essentially flawless, while the MLE does well. The other four estimators all have a marked tendency toward underestimation, particularly CB-SEM.

### 6.3. Multiple reflective composites

Fig. 6.3 gives the results of a simulation with $q = 2$ reflective composites for $X$ and $u = 2$ reflective composites for $Y$. The simulation was structured like that described in Section 6.2, except now $p = r = 21$, $L$ is a $21 \times 2$ matrix and

$$\Psi = 0.9997224 \times \mathrm{cor}(\eta, \xi). \tag{6.2}$$

Details are available in Appendix F.

We see from Fig. 6.3 that the EB-PLS estimator does the best at all sample sizes, while EB-PLS-SEM does well at the larger sample sizes. The Matrixpls estimator from Rönkkö et al. (2016b) underestimates the true correlation $\Psi$ at all displayed sample sizes because it implicitly assumes that $q = u = 1$. The CB-SEM estimator also underestimates its target $\mathrm{cor}(\xi, \eta)$ because it cannot deal with non-diagonal covariance matrices.

## 7. Discussion

The results of this paper are based on the relatively simple path diagram of Fig. 2.1, which follows the presentation style of Wold (1982). As stated in the Introduction, this was done so our context and conclusions might be clear and informative, as it seems to us that many claims in the PLS-SEM v. CB-SEM debate are made without adequate foundations. A question may now arise as to the relevance of our conclusions in complex path models that might involve (i) more than two sets of indicator–construct combinations, (ii) formative relationships between some constructs and their associated indicators and reflective relationships between others, (iii) more constructs than sets of indicators or (iv) complicated inner models. The following overarching conclusions about the role of PLS in path analyses apply regardless of the complexity of the path model. Our conclusions are phrased in terms of a construct $\xi$ and a corresponding vector of indicators $X$. The relationship between the indicators and the construct can be reflective or formative. Our conclusions do not depend on the nature of the inner model.

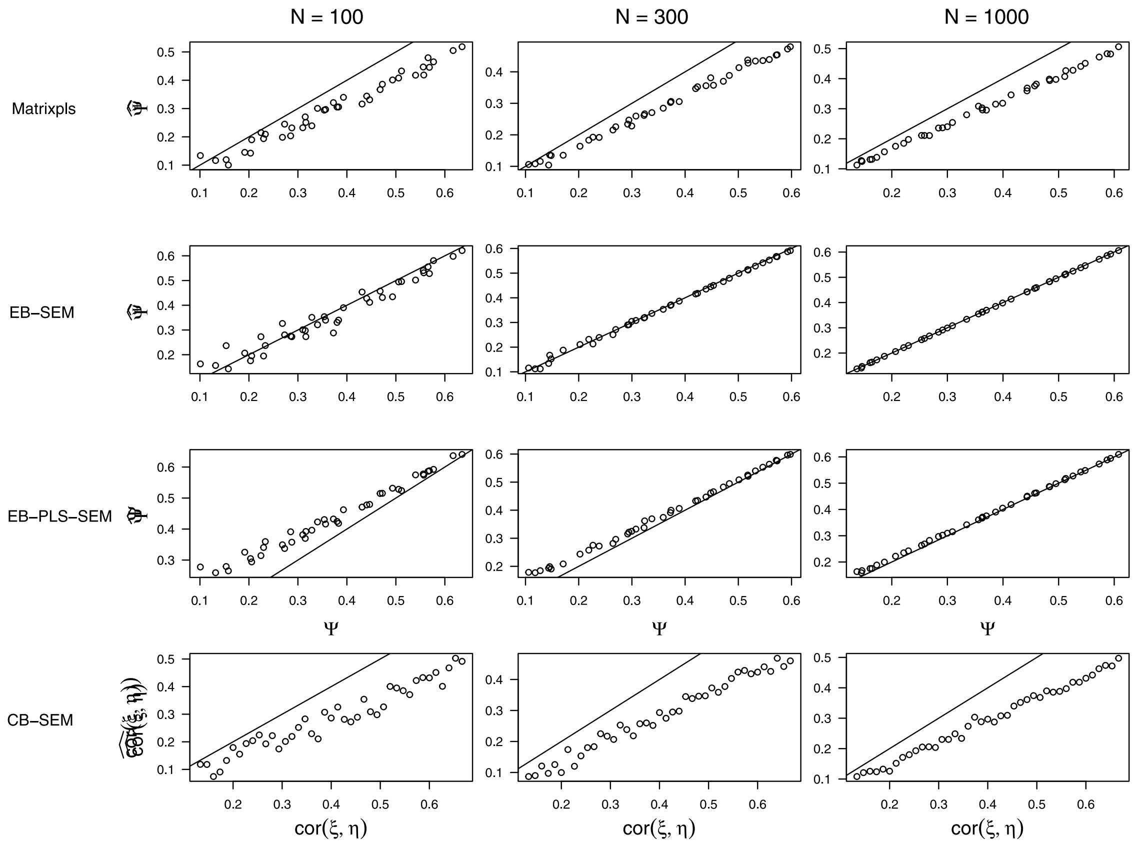


**Fig. 6.3.** Results of simulations with two reflective composites for $X$ and $Y$, $q = u = 2$. The details are given in Appendix F. Horizontal axes give the true correlations; vertical axes give the estimated correlations. The estimators are as described in the legend of Fig. 6.1.

*Path modeling.* At its core, path modeling hinges on the ability of the investigators to identify sets of indicators that are related to the constructs in the manner specified. As discussed in Section 2.3, our view is that an understanding of a construct should not be divorced from the specific indicators selected for its study. This view leads naturally to using $E(\xi \mid X)$ as a means of construct reification. Here PLS-SEM, EB-PLS-SEM, and EB-SEM can have a useful role in reducing the dimension of $X$ without loss of information on $\xi$, which allows $X$ to be replaced by $X_R$ so that $E(\xi \mid X) = E(\xi \mid X_R)$.

On the other hand, if marginal characteristics of the constructs like $\mathrm{cov}(\xi, \eta)$ are of sole interest, then we see PLS as having little or no relevance to the analysis, unless the lower bound $|\Psi| \leq |\mathrm{cov}(\xi, \eta)|$ is useful.

*PLS-SEM.* This method works best when one real composite of the indicators, say $\Phi^T X$, extracts all of the available information from $X$ about $\xi$. That is, once the real composite $\Phi^T X$ is known and fixed, $\xi$ and $X$ are independent or at least uncorrelated. However, we found no rational for adopting this one-composite framework, which we see as tying the hands of PLS-SEM. Envelopes and their descendent PLS methods include methodology for estimating the number of composites needed to extract all of the available information. Expanding the one-composite framework to EB-PLS-SEM has the potential to increase the efficacy of PLS-SEM considerably.

*CB-SEM.* This likelihood-based method requires constraints on covariance matrices for identifiability, as discussed in Section 3.2. We find the common assumption of diagonal covariance matrices to be quite restrictive.

*What do the PLS-SEM weights actually accomplish?* Estimates of the PLS-SEM weight *vector* $\Phi$ are intended to extract the material part of the indicators $X$, giving the composite $\Phi^T X$ and effectively leaving behind the noise. Relaxing the one-component framework, the EB-PLS-SEM weights come in a *matrix* $\Phi$ which is then used to extract multiple composites $\Phi^T X$. These composites hold all of the information about $\xi$ that is available from $X$.

*Is PLS serviceable in small samples.* Yes, in the sense that EB-PLS-SEM does not require $S_X$ and $S_Y$ to be nonsingular.

*Do PLS weights reduce the impact of measurement error?* Yes. By accounting for the material and immaterial parts of $X$, PLS mitigates measurement errors by amounts that depends on the variation of the immaterial part of $X$.

*PLS versus unit-weighted summed composites.* The unit-weighted summed composite is the sum of the individual indicators, $X_{\mathrm{sum}} = \sum_{j=1}^{p} X_j$. The effectiveness of $X_{\mathrm{sum}}$ depends on its relationship to the material composites. If the multiple correlation between $X_{\mathrm{sum}}$ and $X_R$ is high then $X_{\mathrm{sum}}$ may be a useful composite, particularly if the dimension $(q)$ of $X_R$ is small. On the other extreme, if the multiple correlation between $X_{\mathrm{sum}}$ and the immaterial composites $X_{0R}$ is large then $X_{\mathrm{sum}}$ is effectively an immaterial composite and no useful results based on $X_{\mathrm{sum}}$ should be expected. In short, the usefulness of summed composites depends on the anatomy of the paths. In some path analyses they might be quite useful, while useless in others.

*Bias.* We do not see bias as playing a dominant role in the debate over methodology. If the goal is to estimate $\mathrm{cor}(\eta, \xi)$ using CB-SEM then estimation bias may, depending on the sample size, play a notable role in CB-SEM, as maximum likelihood estimators are biased but unbiased asymptotically. If the goal is to estimate $\mathrm{cor}(\eta, \xi)$ using PLS-SEM then structural bias is relevant. But if the goal is to estimate $\Psi$ then structural bias has no special relevance.

*Methodology.* Consider a model with $J$ sets of indicators $X_{jk}$, each influenced reflectively by a single construct, say, $\xi_j$, with 0 mean, $j = 1, \ldots, J, k = 1, \ldots, r_j$. Let $X_j = (X_{j1}, \ldots, X_{jr_j})^T$ denote the vector of indicators for the $j$th construct. All methods discussed here apply to the estimation of $E(\xi_j \mid X_j)$ and of $\Psi_{jk} = \mathrm{cor}\{E(\xi_j \mid X_j), E(\xi_k \mid X_k)\}$: Imbedding the path model shown in Fig. 2.1 in a more complicated model with additional sets of indicator–construct combinations does not impact our conclusions. We can still use canonical correlations to estimate $\Psi_{jk}$, for example. However, a more complicated model with more indicator–construct combinations and a complex inner model may involve additional objectives and methodological issues that require further study. Straightforward adaptation of the methodology can be used to study the regression of $E(\xi_k \mid X_k)$ on other conditional expectations. Complications that necessitate further study arise when the inner model connecting the constructs has constraints or there is not a one-to-one correspondence between constructs and sets of indicators. Complications also arise when there are formative relationships between some constructs and the associated indicators. These complications, as well as allowing for multiple composites of indicators to broaden the scope of PLS-SEM, require additional methodological development, but we see no particular road blocks that might hinder progress. A possible starting point for those developments is sketched in Appendix G.

## CRediT authorship contribution statement

**R. Dennis Cook:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Liliana Forzani:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

## Acknowledgments

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## Appendix A. Multivariate model

The upper and lower path diagrams in Fig. 2.1 can be represented as a common multivariate (multi-response) regression model:

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} \mu_\xi \\ \mu_\eta \end{pmatrix} + \epsilon_{(\xi,\eta)}, \text{ where } \epsilon_{(\xi,\eta)} \sim N_2(0, \Sigma_{(\xi,\eta)}). \tag{A.1}$$

$$Y = \mu_Y + \beta_{Y|\eta}(\eta - \mu_\eta) + \epsilon_{Y|\eta}, \text{ where } \epsilon_{Y|\eta} \sim N_r(0, \Sigma_{Y|\eta}). \tag{A.2}$$

$$X = \mu_X + \beta_{X|\xi}(\xi - \mu_\xi) + \epsilon_{X|\xi}, \text{ where } \epsilon_{X|\xi} \sim N_p(0, \Sigma_{X|\xi}). \tag{A.3}$$

Variables in parentheses like $(X, Y)$, $(X, Y, \eta, \xi)$ and $(\xi, \eta)$ refer to the joint distribution of the variables. We use $\sigma_\xi^2$, $\sigma_{\xi\eta}$ and $\sigma_\eta^2$ to denote the elements of $\Sigma_{(\xi,\eta)} \in \mathbb{R}^{2\times 2}$, and we further assume that $\epsilon_{(\xi,\eta)}$, $\epsilon_{X|\xi}$ and $\epsilon_{Y|\eta}$ are mutually independent so jointly $X$, $Y$, $\xi$ and $\eta$ follow a multivariate normal distribution. We use the notation $\Sigma_{UV}$ to denote the matrix of covariances between the elements of the random vectors $U$ and $V$, and we use $\beta_{U|V} = \Sigma_{UV}\Sigma_V^{-1}$ to indicate the matrix of population coefficients from multi-response regression of $U$ on $V$, where $\Sigma_V = \mathrm{var}(V)$. These are often called loadings in the PLS-SEM literature. The joint distribution of $X$, $Y$, $\xi$ and $\eta$ is given in the following lemma.

**Lemma A.1.** *For model (A.1)–(A.3), we have*

$$\begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix} \sim \begin{pmatrix} \mu_X \\ \mu_Y \\ \mu_\xi \\ \mu_\eta \end{pmatrix} + \epsilon_{(X,Y,\xi,\eta)}$$

*where* $\mathrm{var}(\epsilon_{X,Y,\xi,\eta}) = \Sigma_{(X,Y,\xi,\eta)}$ *with*

$$\Sigma_{(X,Y,\xi,\eta)} = \begin{pmatrix} \Sigma_{X|\xi} + \beta_{X|\xi}\beta_{X|\xi}^T\sigma_\xi^2 & \beta_{X|\xi}\beta_{Y|\eta}^T\sigma_{\xi\eta} & \beta_{X|\xi}\sigma_\xi^2 & \beta_{X|\xi}\sigma_{\xi\eta} \\ \cdots & \Sigma_{Y|\eta} + \beta_{Y|\eta}\beta_{Y|\eta}^T\sigma_\eta^2 & \beta_{Y|\eta}\sigma_{\xi\eta} & \beta_{Y|\eta}\sigma_\eta^2 \\ \cdots & \cdots & \sigma_\xi^2 & \sigma_{\xi\eta} \\ \cdots & \cdots & \cdots & \sigma_\eta^2 \end{pmatrix}. \tag{A.4}$$

**Proof.** The first two elements of the diagonal are direct consequence of the fact that $\mathrm{cov}(Z) = E(\mathrm{cov}(Z|H)) + \mathrm{var}(E(Z|H))$. For $\Sigma_{X\xi}$ and $\Sigma_{Y\eta}$ we use the fact that $\beta_{X|\xi} = \sigma_\xi^{-2}\Sigma_{X\xi}$ and $\beta_{Y|\eta} = \sigma_\eta^{-2}\Sigma_{Y\eta}$. To compute $\Sigma_{Y\eta}$ we use that

$$\begin{aligned} \Sigma_{Y\eta} &= E((Y - \mu_Y)(\eta - \mu_\eta)) \\ &= E_\eta[E((Y - \mu_Y)(\eta - \mu_\eta))|\eta] \\ &= \beta_{Y|\eta}E_\eta(\eta - \mu_\eta)^2 \\ &= \beta_{Y|\eta}\sigma_\eta^2. \end{aligned}$$

Analogously for $\Sigma_{X\xi}$. Now for $\Sigma_{XY}$ we use that

$$\begin{aligned} \Sigma_{YX} &= E((Y - \mu_Y)(X - \mu_X)^T) \\ &= E_{\eta,\xi}[E_{Y,X|(\eta,\xi)}\{(Y - \mu_Y)(X - \mu_X)^T\}] \\ &= \beta_{Y|\eta}E_{\eta,\xi}(\eta - \mu_\eta)(\xi - \mu_\xi)\beta_{X|\xi}^T \\ &= \beta_{Y|\eta}\sigma_{\xi\eta}\beta_{X|\xi}^T. \end{aligned}$$

Now, for $\Sigma_{X\eta}$

$$\begin{aligned} \Sigma_{X\eta} &= E((X - \mu_X)(\eta - \mu_\eta)) \\ &= E_{\eta,\xi}E((X - \mu_X)(\eta - \mu_\eta)|\eta, \xi) \\ &= \beta_{X|\xi}E_{\eta,\xi}E((\xi - \mu_\xi)(\eta - \mu_\eta)|\eta, \xi) \\ &= \beta_{X|\xi}\sigma_{\xi\eta}. \end{aligned}$$

The proof of $\Sigma_{Y\xi}$ is analogous. □

The regression and marginal constraints are related via the variance decompositions $\sigma_\xi^2 = \mathrm{var}\{\mathrm{E}(\xi \mid X)\} + \mathrm{E}\{\mathrm{var}(\xi \mid X)\}$ and $\sigma_\eta^2 = \mathrm{var}\{\mathrm{E}(\eta \mid Y)\} + \mathrm{E}\{\mathrm{var}(\eta \mid Y)\}$. Under the marginal constraints $\sigma_\xi^2 = \sigma_\eta^2 = 1$ and consequently $\mathrm{var}\{\mathrm{E}(\eta \mid Y)\} < 1$ and $\mathrm{var}\{\mathrm{E}(\xi \mid X)\} < 1$.

The use of $\mathrm{cor}\{\mathrm{E}(\eta \mid Y), \mathrm{E}(\xi \mid X)\}$ as a measure of association can be motivated by an appeal to dimension reduction. Under the model (A.1)–(A.3), $X \perp\!\!\!\perp \xi \mid \mathrm{E}(\xi \mid X)$ and $Y \perp\!\!\!\perp \eta \mid \mathrm{E}(\eta \mid Y)$. In consequence, the constructs affect their respective indicators only through the linear combinations given by the conditional means, $\mathrm{E}(\xi \mid X) = \mu_\xi + \Sigma_{X\xi}^T\Sigma_X^{-1}(X - \mu_X)$ and $\mathrm{E}(\eta \mid Y) = \mu_\eta + \Sigma_{Y\eta}^T\Sigma_Y^{-1}(Y - \mu_Y)$.

We introduce the marginal or regression constraints when needed.

## Appendix B. Reduced rank regression

PLS path algorithms gain information on the latent constructs from the multi-response linear regressions of $Y$ on $X$ and $X$ on $Y$. Consequently, it is useful to provide a characterization of these regressions. In the following lemma we state that for the model presented in Appendix A, the regression of $Y$ on $X$ qualifies as a reduced rank regression (RRR). The essential implication of the lemma is that, under the reflective model described in Section 2, the coefficient matrix $\beta_{Y|X}$ in the multi-response linear regression of $Y$ on $X$ must have rank 1 because $\mathrm{cov}(Y, X) = \Sigma_{YX}$ has rank 1. A similar conclusion applies to the regression of $X$ on $Y$. The conclusion that $\Sigma_{YX}$ has rank 1 is due in part to the constructs being real. It is integral, although not necessary, for demonstrating identifiability in Proposition B.1

**Lemma B.1.** *For model presented in Appendix A and without imposing either the regression or marginal constraints, we have* $E(Y \mid X) = \mu_Y + \beta_{Y|X}(X - \mu_X)$, *where* $\beta_{Y|X} = AB^T\Sigma_X^{-1}$, *and* $A \in \mathbb{R}^{r\times 1}$ *and* $B \in \mathbb{R}^{p\times 1}$ *are such that* $\Sigma_{YX} = AB^T$. *Similarly,* $E(X \mid Y) = \mu_X + \beta_{X|Y}(Y - \mu_Y)$ *with* $\beta_{X|Y}BA^T\Sigma_Y^{-1}$.

**Proof.** This is direct consequence of Lemma A.1 since

$$\Sigma_{YX} = \beta_{Y|\eta}\sigma_{\xi\eta}\beta^T_{X|\xi} = \Sigma_{Y\eta}\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}\Sigma_{\xi X},$$

the rank of $\Sigma_{YX}$ is one and the lemma follows. □

It is known from the literature on RRR that the vectors $A$ and $B$ are not identifiable while $AB^T$ is identifiable. However, the subspaces they span, $\mathcal{A} = \mathrm{span}(A) = \mathrm{span}(\Sigma_{YX})$ and $\mathcal{B} = \mathrm{span}(B) = \mathrm{span}(\Sigma_{XY})$ are identifiable. As a consequence of being a reduced rank regression model, we are able to state in Proposition B.1 which parameters are identifiable in the reflective model presented in Section 2. There may not exist serviceable estimators for parameters that are not identifiable. Thus, establishing identifiability is a useful prelude to estimation generally and a necessary prelude to our consideration of maximum likelihood estimation in Appendix C.1.

**Proposition B.1.** *The parameters $\Sigma_X$, $\Sigma_Y$, $\Sigma_{YX} = AB^T$, $\beta_{Y|X}$, $\mu_Y$ and $\mu_X$ are identifiable in the reduced rank model of Lemma B.1, but $A$ and $B$ are not. Additionally, under the regression constraints, the quantities $\sigma_{\xi\eta}/(\sigma_\eta^2\sigma_\xi^2)$, $cor\{E(\xi|X), E(\eta|Y)\}$, $E(\xi|X)$, $E(\eta|Y)$, $\Sigma_{X\xi}$ and $\Sigma_{Y\eta}$ are identifiable in the reflective model except for sign, while $\mathrm{cor}(\eta,\xi)$, $\sigma_\xi^2$, $\sigma_\eta^2$, $\sigma_{\xi\eta}$, $\beta_{X|\xi}$, $\beta_{Y|\eta}$, $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are not identifiable. Moreover,*

$$\begin{aligned}|\mathrm{cor}\{E(\xi|X), E(\eta|Y)\}| = |\sigma_{\xi\eta}|/(\sigma_\eta^2\sigma_\xi^2) &= \mathrm{tr}^{1/2}(\beta_{X|Y}\beta_{Y|X})\\ &= \mathrm{tr}^{1/2}(\Sigma_{XY}\Sigma_Y^{-1}\Sigma_{YX}\Sigma_X^{-1}),\end{aligned} \tag{B.1}$$

*where $|\cdot|$ denotes the absolute value of its argument.*

**Proof.** The first part follows from reduced rank model literature (see for example Cook et al., 2015). Now, using (A.4) and the fact that $\mu_\eta = 0$,

$$E(\eta|Y) = \Sigma_{\eta Y}\Sigma_Y^{-1}(Y-\mu_Y)$$

and therefore

$$\mathrm{var}(E(\eta|Y)) = \Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{\eta Y}^T = 1, \tag{B.2}$$

where we use the hypothesis that $\mathrm{var}(E(\eta|Y)) = 1$ in the last equality. In the same way

$$\mathrm{var}(E(\xi|X)) = \Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{\xi X}^T = 1. \tag{B.3}$$

From Lemma A.1,

$$\Sigma_{YX} = AB^T = \Sigma_{Y\eta}\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}\Sigma_{\xi X}. \tag{B.4}$$

Consequently, $\mathcal{A} = \mathrm{span}(\Sigma_{Y\eta})$ and $\mathcal{B} = \mathrm{span}(\Sigma_{X\xi})$ and thus there are constants $m_\eta$ and $m_\xi$ so that

$$\Sigma_{Y\eta} = Am_\eta \tag{B.5}$$

$$\Sigma_{X\xi} = Bm_\xi. \tag{B.6}$$

(B.4), (B.5) and (B.6) together makes $AB^T = Am_\eta\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}m_\xi B^T$ and therefore

$$m_\eta\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}m_\xi = 1. \tag{B.7}$$

Now, using (B.2) and (B.3) together with (B.5) and (B.6) we have that

$$m_\eta^2 = (A^T\Sigma_Y^{-1}A)^{-1} \tag{B.8}$$

$$m_\xi^2 = (B^T\Sigma_X^{-1}B)^{-1}. \tag{B.9}$$

Taking the positive square roots and plugging into (B.7),

$$\begin{aligned}\sigma_\eta^{-2}|\sigma_{\xi\eta}|\sigma_\xi^{-2} &= (m_\xi m_\eta)^{-1}\\ &= (A^T\Sigma_Y^{-1}A)^{1/2}(B^T\Sigma_X^{-1}B)^{1/2}\\ &= (A^T\Sigma_Y^{-1}AB^T\Sigma_X^{-1}B)^{1/2}\\ &= \mathrm{tr}^{1/2}((AB^T)^T\Sigma_Y^{-1}AB^T\Sigma_X^{-1})\\ &= \mathrm{tr}^{1/2}(\beta_{Y|X}\beta_{X|Y}).\end{aligned} \tag{B.10}$$

From (B.10), $m_\xi^2$ and $m_\eta^2$ are not unique. Nevertheless $m_\eta^2 m_\xi^2$ is unique since any change of $A$ and $B$ should be such $AB^T$ is the same. As a consequence $m_\eta m_\xi$ is unique except for a sign. And therefore $\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}$ is unique except for a sign.

Now, coming back to Eqs. (B.5) and (B.6) we have that

$$\Sigma_{Y\eta} = A(A^T\Sigma_Y^{-1}A)^{-1/2} \tag{B.11}$$

$$\Sigma_{X\xi} = B(B^T\Sigma_X^{-1}B)^{-1/2} \tag{B.12}$$

and again are unique except by a sign. These equations imply that for the purpose of determining $\Sigma_{Y\eta}$ and $\Sigma_{X\xi}$ we can restrict $A$ and $B$ to have length 1.

Now, using again (A.4), the regression constraints, $\mu_\eta = \mu_\xi = 0$ and $\mathrm{var}(E(\xi|X)) = \mathrm{var}(E(\eta|Y)) = 1$, we have

$$\begin{aligned}E(\eta|Y) &= \Sigma_{\eta Y}\Sigma_Y^{-1}(Y-\mu_Y)\\ \sigma_\eta^2 &= E(\mathrm{var}(\eta|Y)) + 1\\ E(\xi|X) &= \Sigma_{\xi X}\Sigma_X^{-1}(X-\mu_X)\\ \sigma_\xi^2 &= E(\mathrm{var}(\xi|X)) + 1.\end{aligned}$$

Recall that we have constrained $\mathrm{var}(E(\eta|Y)) = \Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{Y\eta} = 1$ and $\mathrm{var}(E(\xi|X)) = \Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{X\xi} = 1$. Thus, replacing $\Sigma_{YX} = \Sigma_{Y\eta}\sigma_\eta^{-2}\sigma_{\xi\eta}\sigma_\xi^{-2}\Sigma_{\xi X}$ we have

$$\begin{aligned}|\mathrm{cor}\{E(\eta|Y), E(\xi|X)\}| &= |\mathrm{cov}\{E(\eta|Y), E(\xi|X)\}|\\ &= \Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{YX}\Sigma_X^{-1}\Sigma_{X\xi}\\ &= \Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{Y\eta}\sigma_\eta^{-2}|\sigma_{\xi\eta}|\sigma_\xi^{-2}\Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{X\xi}\\ &= \sigma_\eta^{-2}|\sigma_{\xi\eta}|\sigma_\xi^{-2}.\end{aligned}$$

This gives the first equality in (B.1). The second follows from (B.10), and the third is straightforward.

Now, we will prove that $\sigma_\xi$ and $\sigma_\eta$ are not identifiable. For that, since $\sigma_\eta$ and $\sigma_\xi$ have to be greater than 1 and $\sigma_\eta\sigma_\xi$ should be constant we can change $\xi$ by $C\xi$ and $\eta$ by $C^{-1}\eta$ in such a way that $C\sigma_\xi > 1$ and $C^{-1}\sigma_\eta > 1$. We take any $C$ such that $\sigma_\xi^{-1} < C < \sigma_\eta$ and none of the other parameters change. From this follows that $\sigma_{\xi\eta}$ and $\beta_{X|\xi}$, $\beta_{Y|\eta}$ is not identifiable. It is left to prove that $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are not identifiable. For that we use $\Sigma_X = \Sigma_{X|\xi} + \Sigma_{X\xi}\sigma_\xi^{-2}\Sigma_{\xi X}$. If $\Sigma_{X|\xi}$ is identifiable, since $\Sigma_X$ and $\Sigma_{X\xi}$ are identifiable, $\sigma_\xi$ is identifiable, which is a contradiction since we have already proven that it is not so. The same argument holds for $\Sigma_{Y|\eta}$. □

The following lemma is needed in Appendix D.

**Lemma B.2.** *Under the model of Appendix A and the regression constraints,*

$$\Sigma_X = \Sigma_{X|\xi} + \sigma_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}BB^T \tag{B.13}$$

$$= \Sigma_{X|\xi} + \frac{1}{\sigma_\xi^2 - 1}(B^T\Sigma_{X|\xi}^{-1}B)^{-1}BB^T. \tag{B.14}$$

$$\Sigma_Y = \Sigma_{Y|\eta} + \sigma_\eta^{-2}(A^T\Sigma_Y^{-1}A)^{-1}AA^T \tag{B.15}$$

$$= \Sigma_{Y|\eta} + \frac{1}{\sigma_\eta^2 - 1}(A^T\Sigma_{Y|\eta}^{-1}A)^{-1}AA^T. \tag{B.16}$$

**Proof.** By the covariance formula and the fact that by Proposition B.1 we have $\Sigma_{X\xi} = B(B^T\Sigma_X^{-1}B)^{-1/2}$ and $\Sigma_{Y\eta} = A(A^T\Sigma_Y^{-1}A)^{-1/2}$ from where we get (B.13) and (B.15). Now, taking inverse and using the Woodbury inequality we have

$$B^T\Sigma_X^{-1}B = \sigma_\xi^2\frac{B^T\Sigma_{X|\xi}^{-1}BB^T\Sigma_X^{-1}B}{\sigma_\xi^2 B^T\Sigma_X^{-1}B + B^T\Sigma_{X|\xi}^{-1}B}$$

As a consequence

$$B^T\Sigma_X^{-1}B = B^T\Sigma_{X|\xi}^{-1}B\frac{\sigma_\xi^2 - 1}{\sigma_\xi^2}$$

and (B.14) follows replacing this into (B.13). The proof of (B.16) follows analogously. □

## Appendix C. Estimating $\Psi = |\mathrm{cor}\{E(\xi|X), E(\eta|Y)\}|$

### C.1. Maximum likelihood estimation

Under the model of Appendix A, the maximum likelihood estimator of $\Psi$ makes use of the reduced rank estimators $\widehat{\beta}^{(\mathrm{RR})}_{Y|X}$ and $\widehat{\beta}^{(\mathrm{RR})}_{X|Y}$ of $\beta_{Y|X}$ and $\beta_{X|Y}$. It follows from (B.1) that

$$\widehat{\Psi}_{\mathrm{MLE}} = \mathrm{tr}^{1/2}\left(\widehat{\beta}^{(\mathrm{RR})}_{X|Y}\widehat{\beta}^{(\mathrm{RR})}_{Y|X}\right).$$

To construct $\widehat{\beta}^{(\mathrm{RR})}_{Y|X}$, the rank 1 MLE of $\beta_{Y|X}$ from the reduced rank model in Lemma B.1, we use results from Cook et al. (2015): Construct the sample canonical correlation matrix between $Y$ and $X$, $C_{YX} = S_Y^{-1/2}S_{YX}S_X^{-1/2}$, let $L_1$ and $R_1$ denote the first left and right singular vectors of $C_{YX}$ and let $v_1$ denote its first singular value. Then $\widehat{\beta}^{(\mathrm{RR})}_{Y|X} = S_Y^{1/2}L_1 v_1 R_1^T S_X^{-1/2}$ and $\widehat{\beta}^{(\mathrm{RR})}_{X|Y}$ is obtained in the same way, interchanging the roles of $Y$ and $X$. Then it follows by straightforward algebra that

$$\begin{aligned}\widehat{\Psi}_{\mathrm{MLE}} &= \mathrm{tr}^{1/2}(S_X^{1/2}R_1 v_1 L_1^T S_Y^{-1/2} S_Y^{1/2} L_1 v_1 R_1^T S_X^{-1/2}) \\ &= v_1. \end{aligned} \tag{C.1}$$

Consequently, the first sample canonical correlation coefficient between $X$ and $Y$ is the MLE of $\Psi$.

### C.2. Moment estimation

A straightforward moment estimator $\widehat{\Psi}_{\mathrm{MT}}$ of $\Psi$ can be constructed by simply using moment estimators in (B.1):

$$\widehat{\Psi}_{\mathrm{MT}} = \mathrm{tr}^{1/2}(S_{XY}S_Y^{-1}S_{YX}S_X^{-1}).$$

### C.3. Methods used in social science path analysis

Using Dijkstra (1983), Dijkstra and Henseler (2015a) as guides, the basic estimation method used in the social sciences follows these steps:

1. Scaling the indicators marginally, we work with the scaled variables, $\tilde{X} = \mathrm{diag}^{-1/2}(S_X)(X - \bar{X})$ and $\tilde{Y} = \mathrm{diag}^{-1/2}(S_Y)(Y - \bar{Y})$, where $\mathrm{diag}(\cdot)$ denotes the diagonal matrix with diagonal elements the same as those of the argument.
2. Construct the first eigenvector $\ell_1$ of $S_{\tilde{Y}\tilde{X}}S_{\tilde{X}\tilde{Y}}$ and the first eigenvector $r_1$ of $S_{\tilde{X}\tilde{Y}}S_{\tilde{Y}\tilde{X}}$.
3. Construct the proxy latent variables $\bar{\eta} = r_1^T\tilde{X}$ and $\bar{\xi} = \ell_1^T\tilde{Y}$.
4. Then the estimated covariance and correlation between the proxies $\bar{\eta}$ and $\bar{\xi}$ are

$$\begin{aligned}\mathrm{cov}(\bar{\eta},\bar{\xi}) &= \ell_1^T S_{\tilde{Y}\tilde{X}} r_1 \\ \mathrm{cor}(\bar{\eta},\bar{\xi}) &= \frac{\ell_1^T S_{\tilde{Y}\tilde{X}} r_1}{\{\ell_1^T S_{\tilde{Y}}\ell_1 r_1^T S_{\tilde{X}} r_1\}^{1/2}}.\end{aligned} \tag{C.2}$$

Clearly, this estimator is different from the MLE (C.1). Aside from the scaling, it is the same as the simultaneous PLS method developed by Cook et al. (2023, Section 4.3) when $u = q = 1$.

However, if we standardize jointly as some authors seem to do, then we recover the MLE:

1. Standardizing the indicators jointly, we now work with the standardized variables, $\tilde{X} = S_X^{-1/2}(X - \bar{X})$ and $\tilde{Y} = S_Y^{-1/2}(Y - \bar{Y})$.
2. Construct the first eigenvector $L_1$ of $S_{\tilde{Y}\tilde{X}}S_{\tilde{X}\tilde{Y}}$ and the first eigenvector $R_1$ of $S_{\tilde{X}\tilde{Y}}S_{\tilde{Y}\tilde{X}}$. These eigenvectors are the same as the singular vectors in Appendix C.1, as implied by the notation.
3. Construct the proxy latent variables $\bar{\eta} = R_1^T\tilde{X}$ and $\bar{\xi} = L_1^T\tilde{Y}$.
4. Then the estimated correlation between the proxies $\bar{\eta}$ and $\bar{\xi}$ is

$$\mathrm{cor}(\bar{\eta},\bar{\xi}) = L_1^T S_{\tilde{Y}\tilde{X}} R_1 = v_1$$

Consequently, if joint standardization is used the estimator is the MLE, which is the justification mentioned in Section 3.3.4.

### C.4. Creating proxies generally

The idea of estimating the correlations $\Psi$ and $\mathrm{cor}(\xi,\eta)$ by using proxies seems to be common theme in the path analysis literature. A straightforward rational for creating proxies is to use the regression functions $\mathrm{E}(\xi \mid X)$ and $\mathrm{E}(\eta \mid Y)$ as population proxies for $\xi$ and $\eta$. The population and sample proxies for $\xi$ are then

$$\begin{aligned}\mathrm{E}(\xi \mid X) &= \Sigma_{\xi X}\Sigma_X^{-1}(X - \mu_X) \\ &= (B^T\Sigma_X^{-1}B)^{-1/2}B^T\Sigma_X^{-1}(X - \mu_X) \\ \widehat{\mathrm{E}}(\xi \mid X) &= (\widehat{B}^T S_X^{-1}\widehat{B})^{-1/2}\widehat{B}^T S_X^{-1}(X - \bar{X}),\end{aligned}$$

where $\widehat{B}$ is an estimator of $B$ and we used (B.12) to get the second equality. Similarly, the population and sample proxies for $\eta$ are

$$\begin{aligned}\mathrm{E}(\eta \mid Y) &= \Sigma_{\eta Y}\Sigma_Y^{-1}(Y - \mu_Y) \\ &= (A^T\Sigma_Y^{-1}A)^{-1/2}A^T\Sigma_Y^{-1}(Y - \mu_Y) \\ \widehat{\mathrm{E}}(\eta \mid Y) &= (\widehat{A}^T S_Y^{-1}\widehat{A})^{-1/2}\widehat{A}^T S_Y^{-1}(Y - \bar{Y}),\end{aligned}$$

where $\widehat{A}$ is an estimator of $A$ and we used (B.11) to get the second equality. The absolute population correlation between the proxies is $\Psi$ by definition. Since $\mathrm{var}\{\mathrm{E}(\xi \mid X)\} = \mathrm{var}\{\mathrm{E}(\eta \mid Y)\} = 1$, the sample correlation is

$$\widehat{\Psi}_{\mathrm{prox}} = (\widehat{A}^T S_Y^{-1}\widehat{A})^{-1/2}\widehat{A}^T S_Y^{-1}S_{YX}S_X^{-1}\widehat{B}(\widehat{B}^T S_X^{-1}\widehat{B})^{-1/2}.$$

If we standardize $Y \leftarrow S_Y^{-1/2}(Y - \bar{Y})$ and $X \leftarrow S_X^{-1/2}(X - \bar{X})$ so the sample covariance matrices for the standardized variables are the identity and use the first singular vectors of $C_{X,Y}$ as $\widehat{A}$ and $\widehat{B}$ then in the standardized scale the proxies reduce to $\widehat{\mathrm{E}}(\eta \mid Y) = \widehat{A}^T Y$ and $\widehat{\mathrm{E}}(\xi \mid X) = \widehat{B}^T X$, and the proxy correlation reduces to the maximum likelihood estimator $\widehat{\Psi}_{\mathrm{prox}} = \widehat{\Psi}_{\mathrm{MLE}}$. Otherwise, finding useful proxies via scaling and choices of $\widehat{A}$ and $\widehat{B}$ may take additional goals and justification.

## Appendix D. SEM

As was stated in Proposition B.1, the quantity $\mathrm{cor}(\eta,\xi)$ is not identifiable without further assumptions. We give now sufficient conditions to have identification of $\mathrm{cor}(\xi,\eta)$. The conditions needed are related to the identification of $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$. As seen in Lemma B.2,

$$\Sigma_X = \Sigma_{X|\xi} + \sigma_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}BB^T \tag{D.1}$$

$$\Sigma_Y = \Sigma_{Y|\eta} + \sigma_\eta^{-2}(A^T\Sigma_Y^{-1}A)^{-1}AA^T. \tag{D.2}$$

The terms $(B^T\Sigma_X^{-1}B)^{-1}BB^T$ and $(A^T\Sigma_Y^{-1}A)^{-1}AA^T$ in (D.1) and (D.2) are identifiable. Since the goal of SEM is to estimate $\mathrm{cor}(\xi,\eta)$ and since we know from Proposition B.1 that

$$|\mathrm{cor}\{E(\xi \mid X), E(\eta \mid Y)\}| = |\sigma_{\xi\eta}|/(\sigma_\xi^2\sigma_\eta^2) \tag{D.3}$$

is identifiable, we will be able to identify $|\mathrm{cor}(\xi,\eta)|$ if we can identify $\sigma_\xi^2$ and $\sigma_\eta^2$. From (D.1) and (D.2) that is equivalent to identifying $\Sigma_{Y|\eta}$ and $\Sigma_{X|\eta}$. We show in Proposition D.1 that (a) if $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are identifiable then $\sigma_\xi^2$, $\sigma_\eta^2$, $|\sigma_{\xi\eta}|$, $\beta_{X|\xi}$, $\beta_{Y|\eta}$ are identifiable and that (b) $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are identifiable if and only if $\sigma_\xi^2$, $\sigma_\eta^2$ are so.

**Proposition D.1.** *Assume the regression constraints. If $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are identifiable then $\sigma_\xi^2$, $\sigma_\eta^2$, $|\sigma_{\xi\eta}|$, $\beta_{X|\xi}$, $\beta_{Y|\eta}$ are identifiable, and*

$$\mathrm{cor}(\eta,\xi) = \mathrm{cor}\{E(\eta|Y), E(\xi|X)\}\sigma_\xi\sigma_\eta \tag{D.4}$$

$$\sigma_\xi^2 = \frac{H_\xi}{H_\xi - 1} \tag{D.5}$$

$$\sigma_\eta^2 = \frac{H_\eta}{H_\eta - 1} \tag{D.6}$$

$$\begin{aligned}H_\xi &= \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi} = B^T\Sigma_{X|\xi}^{-1}B/B^T\Sigma_X^{-1}B \\ H_\eta &= \Sigma_{\eta Y}\Sigma_{Y|\eta}^{-1}\Sigma_{Y\eta} = A^T\Sigma_{Y|\eta}^{-1}A/A^T\Sigma_Y^{-1}A\end{aligned}$$

*where $A$ and $B$ are as defined in Lemma B.1. Moreover, $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are identifiable if and only if $\sigma_\xi^2$, $\sigma_\eta^2$ are so.*

**Proof.** By (B.13)–(B.16) and using Proposition B.1 we have that $\sigma_\xi$ and $\sigma_\eta$ are identifiable and as a consequence the rest of the parameters are identifiable. Now, to prove (D.5) let us use the formula

$$\Sigma_X = \Sigma_{X|\xi} + \Sigma_{X\xi}\sigma_\xi^{-2}\Sigma_{\xi X}.$$

Using Woodbury inequality

$$\Sigma_X^{-1} = \Sigma_{X|\xi}^{-1} - \Sigma_{X|\xi}^{-1}\Sigma_{X\xi}(\sigma_\xi^2 + \Sigma_{\xi X}\Sigma_{X\xi}^{-1}\Sigma_{X\xi})^{-1}\Sigma_{\xi X}\Sigma_{X|\xi}^{-1}. \tag{D.7}$$

Using the fact that $\Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{X\xi} = 1$ proven in Proposition B.1 and multiplying to the left and to the right of (D.7) by $\Sigma_{\xi X}$ and $\Sigma_{X\xi}$ we have

$$\begin{aligned} 1 &= \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi} - \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi}(\sigma_\xi^2 + \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi})^{-1}\Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi} \\ &= H_\xi - H_\xi(\sigma_\xi^2 + H_\xi)^{-1}H_\xi \\ &= \frac{H_\xi\sigma_\xi^2}{\sigma_\xi^2 + H_\xi} \end{aligned}$$

from where we get (D.5). Analogously we get (D.6). □

The next proposition and its corollary gives conditions that are sufficient to ensure identifiability. Consistent with the notation used in Section 2.1, let $M_{ij}$ denote the $ij$-th element of the matrix $M$ and $V_i$ the $i$th element of the vector $V$.

**Proposition D.2.** *Under the regression constraints, (I) if $\Sigma_{X|\xi}$ contains an off-diagonal element that is known to be zero, say $(\Sigma_{X|\xi})_{ij} = 0$, and if $B_iB_j \neq 0$ then $\Sigma_{X|\xi}$ and $\sigma_\xi^2$ are identifiable. (II) if $\Sigma_{Y|\eta}$ contains an off-diagonal element that is known to be zero, say $(\Sigma_{Y|\eta})_{ij} = 0$, and if $(A)_i(A)_j \neq 0$ then $\Sigma_{Y|\eta}$ and $\sigma_\eta^2$ are identifiable.*

*If $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are diagonal matrices and at least two components of $B$ and at least two components of $A$ are nonzero then $\Sigma_{X|\xi}$, $\sigma_\xi^2$, $\Sigma_{Y|\eta}$ and $\sigma_\eta^2$ are identifiable.*

**Proof.** Identification of $\Sigma_{X|\xi}$ means that if we have two different matrices $\Sigma_X$ and $\tilde{\Sigma}_X$ satisfying (D.1) then we should conclude that $\Sigma_{X|\xi} = \tilde{\Sigma}_{X|\xi}$ and that $\sigma_\xi^2 = \tilde{\sigma}_\xi^2$. The same logic applies to $\Sigma_{Y|\eta}$ and $\sigma_\eta^2$. More specifically, from (D.1) the equality

$$\Sigma_{X|\xi} + \sigma_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}BB^T = \tilde{\Sigma}_{X|\xi} + \tilde{\sigma}_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}BB^T \tag{D.8}$$

must imply that $\Sigma_{X|\xi} = \tilde{\Sigma}_{X|\xi}$ and $\sigma_\xi^2 = \tilde{\sigma}_\xi^2$. If $\sigma_\xi^2$ is identifiable, so $\sigma_\xi^2 = \tilde{\sigma}_\xi^2$, then (D.8) implies $\Sigma_{X|\xi} = \tilde{\Sigma}_{X|\xi}$ since $(B^T\Sigma_X^{-1}B)^{-1}BB^T$ is identifiable. Similarly, if $\Sigma_{X|\xi}$ is identifiable, so $\Sigma_{X|\xi} = \tilde{\Sigma}_{X|\xi}$, then (D.8) implies that $\sigma_\xi^2 = \tilde{\sigma}_\xi^2$ and thus that $\sigma_\xi^2$ is identifiable.

Now, assume that elements $(i,j)$ and $(j,i)$ of $\Sigma_{X|\xi}$ and $\tilde{\Sigma}_{X|\xi}$ are known to be 0 and let $e_k$ denote the $p \times 1$ vector with a 1 in position $k$ and 0's elsewhere. Then multiplying (D.8) on the left by $e_i$ and on the right by $e_j$ gives

$$\sigma_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}B_iB_j = \tilde{\sigma}_\xi^{-2}(B^T\Sigma_X^{-1}B)^{-1}B_iB_j.$$

Since $(B^T\Sigma_X^{-1}B)^{-1}BB^T$ is identifiable and $B_iB_j \neq 0$, this implies that $\sigma_\xi^{-2} = \tilde{\sigma}_\xi^{-2}$ and thus $\sigma_\xi^2$ is identifiable, which implies that $\Sigma_{X|\xi}$ is identifiable. □

**Corollary D.1.** *Under the regression constraints, if $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ are diagonal matrices and if $A$ and $B$ each contain at least two non-zero elements then $\Sigma_{Y|\eta}$, $\Sigma_{X|\xi}$, $\sigma_\xi^2$ and $\sigma_\eta^2$ are identifiable.*

The usual assumption in the SEM model is that $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are diagonal matrices (eg. Henseler et al., 2014). We see from (D.3) and Corollary D.1 that this assumption along with the regression constraints is sufficient to guarantee that $|\mathrm{cor}(\xi,\eta)|$ is identifiable provided $B$ and $A$ contain at least two non-zero elements. However, from Proposition D.2 we also see that is not necessary for $\Sigma_{Y|\eta}$ and $\Sigma_{X|\xi}$ to be diagonal.

In full, the usual SEM model requires that $\Sigma_{Y|\eta} = D_{Y|\eta}$ and $\Sigma_{X|\xi} = D_{X|\xi}$ are diagonal matrices, $D_{Y|\eta}$ and $D_{X|\xi}$, and it adopts the marginal constraints instead of the regression constraints. By Proposition D.2, our ability to identify parameters is unaffected by the constraints adopted. However, we need also to be sure that the constraints do not alter the meaning of $\mathrm{cor}(\xi,\eta)$ as it relates to other parameters in the model.

**Lemma D.1.** *In the SEM model, $\mathrm{cor}(\xi,\eta)$ is the same under the marginal constraints $\sigma_\xi^2 = \sigma_\eta^2 = 1$ as it is under the regression constraints $\mathrm{var}\{E(\xi|X)\} = \mathrm{var}\{E(\eta|Y)\} = 1$.*

**Proof.** If we have $\sigma_\xi^2 = \sigma_\eta^2 = 1$ and we define $\tilde{\xi} = c_\xi\xi$, $\tilde{\eta} = c_\eta\xi$ with $c_\xi = [\mathrm{var}\{E(\xi|X)\}]^{-1/2}$ (we can defined because $\mathrm{var}E(\xi|X)$ is identifiable) we have that $\mathrm{cor}(\tilde{\eta},\tilde{\xi}) = \mathrm{cor}(\xi,\eta)$. Now, if $\mathrm{var}\{E(\eta|Y)\} = \mathrm{var}\{E(\xi|X)\} = 1$ and $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are identifiable we can identify $\sigma_\xi$ and $\sigma_\eta$ and we could define $\tilde{\xi} = c_\xi\xi$, $\tilde{\eta} = c_\eta\xi$ with $c_\xi = \{\sigma_\xi\}^{-1/2}$ and $c_\eta = \{\sigma_\eta\}^{-1/2}$ since $c_\xi$ and $c_\eta$ are identifiable. □

Now, to estimate the identifiable parameters with the regression or marginal constraints, we take the joint distribution of $(X,Y)$ and insert the conditions imposed. The maximum likelihood estimators can then be found under normality using the implied likelihoods which can be obtained from the results of Proposition D.3.

**Proposition D.3.** *The SEM models under the marginal and regression constraints are*

- *Marginal constraints,*

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} + \epsilon_{X,Y}$$

*where*

$$\mathrm{var}(\epsilon_{X,Y}) := \Sigma_{X,Y} = \begin{pmatrix} D_{X|\xi} + c^2BB^T & BA^T \\ AB^T & D_{Y|\eta} + d^2AA^T \end{pmatrix}. \tag{D.9}$$

*and $c^2d^2\mathrm{cor}^2(\eta,\xi) = 1$.*

- *Regression constraints*

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} + \epsilon_{X,Y}$$

*From Lemma B.1 and (B.14),*

$$\Sigma_{X,Y} = \begin{pmatrix} D_{X|\xi} + (B^TD_{X|\xi}^{-1}B)^{-1}BB^T/(\sigma_\xi^2 - 1) & BA^T \\ AB^T & D_{Y|\eta} + (A^TD_{Y|\eta}^{-1}A)^{-1}AA^T/(\sigma_\eta^2 - 1) \end{pmatrix}.$$

*where $\mathrm{cor}^2(\eta,\xi)(\sigma_\eta^2 - 1)^{-1}(A^TD_{Y|\eta}^{-1}A)^{-1}(\sigma_\xi^2 - 1)^{-1}(B^TD_{X|\xi}^{-1}B)^{-1} = 1.$*

**Proof.** For (D.9) we only need to prove $\Sigma_X = D_{X|\xi} + BB^Tc^2$ and $\Sigma_Y = D_{Y|\eta} + AA^Td^2$ with $c^2d^2\mathrm{cor}(\eta,\xi) = 1$.

Now, $\Sigma_X = D_{X|\xi} + \Sigma_{X\xi}\Sigma_{\xi X}$ and $\Sigma_Y = D_{Y|\eta} + \Sigma_{Y\eta}\Sigma_{\eta Y}$. And since $\Sigma_{YX} = AB^T = \Sigma_{Y\eta}\mathrm{cor}(\eta,\xi)\Sigma_{\xi X}$ we have that $\Sigma_{Y\eta} = Ad$ and $\Sigma_{X\xi} = Bc$ with $dc \times \mathrm{cor}(\eta,\xi) = 1$ from what follows (D.9). Then the conclusion for the regression constraints follows from the proof of Lemma B.2. □

From this we conclude that the same maximization problem arises under either sets of constraints.

## Appendix E. Bias

We understand that the notion of bias in the literature on path analysis refers to the difference between measuring the association via $\mathrm{cor}\{E(\xi|X), E(\eta|Y)\}$ with the regression constraints and via $\mathrm{cor}(\xi,\eta)$ with the marginal constraints. To establish this connection, we let $\xi$ and $\eta$ denote the latent variables under the marginal constraints, so $\mathrm{var}(\xi) = \mathrm{var}(\eta) = 1$, and let $\tilde{\xi} = \xi\mathrm{var}^{-1/2}\{E(\xi \mid X)\}$ and $\tilde{\eta} = \eta\mathrm{var}^{-1/2}\{E(\eta \mid Y)\}$ denote the corresponding latent variables with the regression constraints, so $\mathrm{var}\{E(\tilde{\xi} \mid X)\} = \mathrm{var}\{E(\tilde{\eta} \mid Y)\} = 1$. Then

**Proposition E.1.**

$\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\} = [\mathrm{var}\{E(\xi \mid X)\}\mathrm{var}\{E(\eta \mid Y)\}]^{1/2}\mathrm{cor}(\xi,\eta)$.

**Proof.**

$$\begin{aligned}\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\} &= [\mathrm{var}\{E(\xi \mid X)\}\mathrm{var}\{E(\eta \mid Y)\}]^{-1/2} \\ &\quad\times \mathrm{cov}\{E(\xi \mid X), E(\eta \mid Y)\} \\ &= [\mathrm{var}\{E(\xi \mid X)\}\mathrm{var}\{E(\eta \mid Y)\}]^{1/2}\mathrm{cor}(\xi,\eta).\end{aligned}$$

Now,

$$\begin{aligned}\mathrm{cov}\{E(\xi \mid X), E(\eta \mid Y)\} &= \mathrm{cov}(\Sigma_{\xi X}\Sigma_X^{-1}X, \Sigma_{\eta Y}\Sigma_Y^{-1}Y) \\ &= \Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{XY}\Sigma_Y^{-1}\Sigma_{Y\eta}.\end{aligned}$$

From (A.4), $\Sigma_{XY} = \beta_{X|\xi}\beta_{Y|\eta}\sigma_{\xi\eta}$. Because $\sigma_\xi^2 = \sigma_\eta^2 = 1$ this can be re-expressed as $\Sigma_{XY} = \Sigma_{X\xi}\Sigma_{\eta Y}\sigma_{\xi\eta} = \Sigma_{X\xi}\Sigma_{\eta Y}\mathrm{cor}(\xi,\eta)$ and so

$$\begin{aligned}\mathrm{cov}\{E(\xi \mid X), E(\eta \mid Y)\} &= \Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{X\xi}\Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{Y\eta}\mathrm{cor}(\xi,\eta) \\ &= [\mathrm{var}\{E(\xi \mid X)\}\mathrm{var}\{E(\eta \mid Y)\}]\mathrm{cor}(\xi,\eta).\end{aligned}$$

Multiplying both sides by $[\mathrm{var}\{E(\xi \mid X)\}\mathrm{var}\{E(\eta \mid Y)\}]^{-1/2}$ completes the justification.

To perhaps aid intuition, we can see the result in another way by using the Woodbury identity to invert $\Sigma_X = \Sigma_{X\xi}\Sigma_{\xi X} + \Sigma_{X|\xi}$ we have

$$\begin{aligned}\Sigma_{\xi X}\Sigma_X^{-1}\Sigma_{X\xi} &= \Sigma_{\xi X}(\Sigma_{X\xi}\Sigma_{\xi X} + \Sigma_{X|\xi})^{-1}\Sigma_{X\xi} \\ &= \frac{\Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi}}{1 + \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi}}.\end{aligned}$$

Similarly,

$$\Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{Y\eta} = \frac{\Sigma_{\eta Y}\Sigma_{Y|\eta}^{-1}\Sigma_{Y\eta}}{1 + \Sigma_{\eta Y}\Sigma_{Y|\eta}^{-1}\Sigma_{Y\eta}},$$

and thus

$$\begin{aligned}&\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\} \\ &= \left[\frac{\Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi}}{1 + \Sigma_{\xi X}\Sigma_{X|\xi}^{-1}\Sigma_{X\xi}}\,\frac{\Sigma_{\eta Y}\Sigma_{Y|\eta}^{-1}\Sigma_{Y\eta}}{1 + \Sigma_{\eta Y}\Sigma_{Y|\eta}^{-1}\Sigma_{Y\eta}}\right]^{1/2}\mathrm{cor}(\xi,\eta).\end{aligned}$$

This form again shows that $\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\} \le \mathrm{cor}(\xi,\eta)$ and provides a more detailed expression of their ratio. □

To interpret Proposition E.1 we use the decomposition

$$\mathrm{var}(\xi) = 1 = E\{\mathrm{var}(\xi \mid X)\} + \mathrm{var}\{E(\xi \mid X)\} = \mathrm{var}(\xi \mid X) + \mathrm{var}\{E(\xi \mid X)\},$$

where the final equality holds because under normality $\mathrm{var}(\xi \mid X)$ is non-stochastic. It follows from this representation that $\mathrm{var}\{E(\xi \mid X)\} \le 1$ and $\mathrm{var}\{E(\eta \mid Y)\} \le 1$. In consequence, $|\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\}|$ is a lower bound on $|\mathrm{cor}(\xi,\eta)|$:

$$|\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\}| \le |\mathrm{cor}(\xi,\eta)|. \tag{E.1}$$

The term $\mathrm{var}(\xi \mid X)$ represents how well $X$ predicts $\xi$. If this term is small relative to $\mathrm{var}(\xi) = 1$, then $\mathrm{var}\{E(\xi \mid X)\}$ will be close to 1, and in consequence

$$|\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\}| \approx [\mathrm{var}\{E(\eta \mid Y)\}]^{1/2}|\mathrm{cor}(\xi,\eta)|.$$

The conditional variance $\mathrm{var}(\xi \mid X)$ will be small when the signal is strong. This will happen with few highly informative predictors. It will also happen as number of informative indicators increases, a scenario that is referred to as an *abundant* regression in statistics (Cook et al., 2012, 2013). Repeating this argument for $\mathrm{var}(\eta \mid Y)$ we see that

$$|\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\}| \approx |\mathrm{cor}(\xi,\eta)|.$$

when $X$ and $Y$ are such that $\mathrm{var}(\xi \mid X)$ and $\mathrm{var}(\eta \mid Y)$ are both small relative to 1, which is often referred to as consistency-at-large in the literature on path analysis (Hui & Wold, 1982). Sufficient conditions for this to occur are that (a) the eigenvalues of $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are bounded away from 0 and $\infty$ as $p, r \to \infty$ and (b) $\Sigma_{X\xi}^T\Sigma_{X\xi}$ and $\Sigma_{Y\eta}^T\Sigma_{Y\eta}$ both diverge as $p, r \to \infty$. This result seems to align well with the argument given at the end of Section 2: $\mathrm{cor}\{E(\tilde{\xi} \mid X), E(\tilde{\eta} \mid Y)\}$ and $\mathrm{cor}(\xi,\eta)$ agree when $X$ and $Y$ provide (nearly) exhaustive information about $\xi$ and $\eta$. The crucial SEM assumption that $\Sigma_{X|\xi}$ and $\Sigma_{Y|\eta}$ are diagonal matrices plays no role in this conclusion.

## Appendix F. Envelopes and PLS

In this section we provide a few details to compliment our discussion of PLS in Section 4.

Let $X_R = \Phi^T X$ and $Y_R = \Gamma^T Y$ denote the vectors of reflective composites, where $\Phi$ is a $p \times q$ matrix with $q \le p$ and $\Gamma$ is an $r \times u$ matrix with $u \le r$. Let $(\Phi, \Phi_0)$ be a full rank $p \times p$ matrix with $\Phi^T\Phi_0 = 0$, and let $(\Gamma, \Gamma_0)$ be a full rank $r \times r$ matrix with $\Gamma^T\Gamma_0 = 0$. Then the complementary composites are $X_{0R} = \Phi_0^T X$ and $Y_{0R} = \Gamma_0^T Y$ are the parts of $X$ and $Y$ that carry no information about $\xi$ and $\eta$. Introducing these constructions into the model of Appendix A gives the following envelope model for the multivariate regression of $Y$ on $X$ (Cook & Zhang, 2015):

$$\begin{aligned}Y &= \alpha + \Gamma\beta_{\Gamma^T Y|\Phi^T X}\Phi^T X + \varepsilon \\ \Sigma_X &= \Phi\Delta\Phi^T + \Phi_0\Delta_0\Phi_0^T \\ \Sigma_{Y|X} &= \Gamma\Omega\Gamma^T + \Gamma_0\Omega\Gamma_0^T,\end{aligned}$$

where $\mathrm{var}(X_R) = \Omega$, $\mathrm{var}(X_{0R}) = \Omega_0$, $\mathrm{var}(Y_R) = \Delta$, $\mathrm{var}(Y_{0R}) = \Delta_0$ and $\beta_{\Gamma^T Y|\Phi^T X}$ is the matrix of regression coefficients for the multivariate regression of $\Gamma^T Y$ on $\Phi^T X$. The matrices $\Gamma$ and $\Phi$ are not identifiable in this model, but the subspaces spanned by their columns are identifiable, which is all that is necessary for estimation of $\Phi$. The MLEs of the parameters in this model were studied by Cook and Zhang (2015), with $q$ and $u$ estimated via an information criterion. The PLS estimators can be constructed by applying a standard PLS regression algorithm. In either case, $q$ and $u$ can also be estimated by cross validation. See Cook (2018) for additional discussion.

For the simulation of Section 6.3, let $1_k$ be a $k \times 1$ vector of ones, let $L_1 = (8, -0.7, 60, 1_9^T, -1_9^T)^T$, $L_2 = (1, 0.3, -0.59/60, -1_9^T, 1_9^T)^T$ and $L = (L_1, L_2)$. The conditional means and variances were generated as $\mathrm{E}(X \mid \xi) = 0.1(L_1 + 0.9L_2)\xi$, $\Sigma_{X|\xi} = 5L(L^T L)^{-1}L^T + 0.1L_0L_0^T$, $E(Y \mid \eta) = 0.1(L_1 + 0.9L_2)\eta$. , $\Sigma_{Y|\eta} = 5L(L^T L)^{-1}L^T + 0.1L_0L_0^T$. From this structure we have, $\Sigma_{Y,\eta} = \Sigma_{X,\xi} = 0.1(L_1 + L_2)$, and

$$\Sigma_{Y,X} = 0.01(L_1 + L_2)(L_1 + L_2)^T = 0.01L1_2 1_2^T L^T.$$

It follows from this structure that $\Phi = \Gamma = L(L^T L)^{-1/2}$ and so $q = u = 2$.

## Appendix G. Extensions

The indicators come in blocks $Y_j \in \mathbb{R}^{r_j}$, $j = 1, \dots, J$, with each block of indicators a reflection of a real latent variable $\eta_j$. There can be more latent variables than blocks, but here we keep them in one-to-one correspondence. Each block is represented reflectively in a path diagram as one of the two blocks in Fig. 2.1. Let $r = \sum_j r_j$,

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_J \end{pmatrix}_{r\times 1}, \quad \epsilon_{Y|\eta} = \begin{pmatrix} \epsilon_{Y_1|\eta_1} \\ \epsilon_{Y_2|\eta_2} \\ \vdots \\ \epsilon_{Y_J|\eta_J} \end{pmatrix}_{r\times 1}, \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_J \end{pmatrix}_{J\times 1}.$$

Then the models for $Y_j$ and $Y$ can be represented following the models in Appendix A:

$$\begin{aligned}Y_j &= \mu_{Y_j} + \beta_{Y_j|\eta_j}\eta_j + \epsilon_{Y_j|\eta_j}, \text{ where } \epsilon_{Y_j|\eta_j} \sim N_r(0, \Sigma_{Y_j|\eta_j}), \\ &\quad j = 1, \dots, J,\end{aligned} \tag{G.1}$$

$$Y = \mu_Y + \beta_{Y|\eta}\eta + \epsilon_{Y|\eta}, \text{ where } \epsilon_{Y|\eta} \sim N_r(0, \Sigma_{Y|\eta}). \tag{G.2}$$

The coefficient matrix $\beta_{Y|\eta}$ is block diagonal

$$\beta_{Y|\eta} = \mathrm{bdiag}(\beta_{Y_1|\eta_1}, \ldots, \beta_{Y_J|\eta_J}).$$

The latent variables are marginally normally distributed $\eta_j \sim N(0, \sigma^2_{\eta_j})$, where the mean is taken to be 0 following the constraints described in Section 2 (Full regression or marginal constraints are not yet imposed.) Extending (A.2) and (A.3), we require the error vectors $\epsilon_{Y_j|\eta_j}$ to be mutually independent so $\Sigma_{Y|\eta}$ is block diagonal

$$\Sigma_{Y|\eta} = \mathrm{bdiag}(\Sigma_{Y_1|\eta_1}, \ldots, \Sigma_{Y_J|\eta_J}).$$

From this setup we get the multi-construct version of (A.4):

$$\begin{aligned}
\Sigma_{(Y,\eta)} &= \begin{pmatrix} \Sigma_Y & \Sigma_{Y\eta} \\ \Sigma_{\eta Y} & \Sigma_\eta \end{pmatrix} \\
&= \begin{pmatrix} \beta_{Y|\eta}\Sigma_\eta\beta^T_{Y|\eta} + \Sigma_{Y|\eta} & \beta_{Y|\eta}\Sigma_\eta \\ \Sigma_\eta\beta^T_{Y|\eta} & \Sigma_\eta \end{pmatrix} \\
\mathrm{var}(E(Y \mid \eta)) &= \beta_{Y|\eta}\Sigma_\eta\beta^T_{Y|\eta} \\
E(\mathrm{var}(Y \mid \eta)) &= \Sigma_{Y|\eta}.
\end{aligned}$$

Additionally, we get

$$\begin{aligned}
E(\eta \mid Y) &= \Sigma_{\eta Y}\Sigma_Y^{-1}(Y - \mu_Y) \\
\mathrm{var}(\eta \mid Y) &= \Sigma_\eta - \Sigma_{\eta Y}\Sigma_Y^{-1}\Sigma_{Y\eta}.
\end{aligned}$$

In contrast to settings with just $J = 2$ latent variables, as represented in Fig. 2.1, where there are many ways in which the latent variables can related to each other. Their interrelationships among the latents are usually stipulated as the marginal model

$$\eta = B\eta + \zeta, \tag{G.3}$$

where $\zeta \sim N(0, \Sigma_\zeta)$, the elements $b_{jk}$ of the $J \times J$ matrix $B$ have $b_{kk} = 0$, $k = 1, \ldots, J$, and it is required that $I - B$ be nonsingular. Solving for $\eta$,

$$\eta = (I - B)^{-1}\zeta \text{ and } \mathrm{var}(\eta) = (I - B)^{-1}\Sigma_\zeta(I - B)^{-1}.$$

Substituting into the model for $Y$ we have

$$Y = \mu_Y + \beta_{Y|\eta}(I - B)^{-1}\zeta + \epsilon_{Y|\eta}.$$

The variance–covariance matrix of $(Y, \eta)$ is computed just as (A.4) was computed. For instance, the variance–covariance matrix of $(Y_k, Y_j, \eta_h, \eta_i)$ is the same as (A.4) with the various quantities now subscripted.

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